applied sciences

MDPI

Article

Optimizing Large Language Models: A Deep Dive into Effective Prompt Engineering Techniques

Minjun Son [1], Yun-Jae Won [2] and Sungjin Lee [3]*

[1] Department of Metabiohealth, Sungkyunkwan University, Suwon 16419, Republic of Korea; mj.son14820@gmail.com
[2] Korea Electronics Technology Institute, Seongnam-si 13488, Republic of Korea; yjwon@keti.re.kr
[3] Department of Smart Automotive, Soonchunhyang University, Asan-si 31538, Republic of Korea
* Correspondence: sungjinlee@sch.ac.kr

Abstract: Recent advancements in Natural Language Processing (NLP) technologies have been driven at an unprecedented pace by the development of Large Language Models (LLMs). However, challenges remain, such as generating responses that are misaligned with the intent of the question or producing incorrect answers. This paper analyzes various Prompt Engineering techniques for large-scale language models and identifies methods that can optimize response performance across different datasets without the need for extensive retraining or fine-tuning. In particular, we examine prominent Prompt Engineering techniques including In-Context Learning (ICL), Chain of Thought (CoT), Retrieval-Augmented Generation (RAG), Step-by-Step Reasoning (SSR), and Tree of Thought (ToT), and we apply these techniques to leading LLMs such as Gemma2, LlaMA3, and Mistral. The performance of these models was evaluated using the AI2 Reasoning Challenge (ARC), HellaSwag, Massive Multitask Language Understanding (MMLU), TruthfulQA, Winogrande, and Grade School Math (GSM8k) datasets across metrics such as BLEU, ROUGE, METEOR, BLEURT, and BERTScore. The experimental results indicate that the most suitable Prompt Engineering technique can vary depending on the characteristics of each dataset. Specifically, for datasets emphasizing mathematical and logical reasoning, Prompt Engineering strategies centered around CoT, SSR, and ToT were found to be advantageous. For datasets focusing on natural language understanding, ICL-centric strategies were more effective, while RAG-based strategies were beneficial for datasets where factual accuracy is crucial. However, it was also observed that the optimal combination of Prompt Engineering techniques could differ depending on the specific LLM, indicating that fine-tuning the Prompt Engineering approach to the model and dataset is essential for achieving the best performance. The findings indicate that as LLMs become more advanced, their reliance on Prompt Engineering (PE) techniques diminishes, yet the magnitude of their performance improvement when PE strategies are applied increases. Furthermore, these advanced models tend to depend less on ICL techniques while exhibiting a greater reliance on RAG strategies. It is also evident that implementing RAG with PE-based preprocessing yields superior performance enhancements compared to the mere application of RAG on raw data.

Keywords: large language model; prompt engineering; in-context learning; chain of thought; retrieval-augmented generation; step-by-step reasoning; tree of thought

check for updates

Academic Editor: Jose María Alvarez Rodríguez

Received: 23 November 2024
Revised: 26 December 2024
Accepted: 24 January 2025
Published: 30 January 2025

Citation: Son, M.; Won, Y.-J.; Lee, S. Optimizing Large Language Models: A Deep Dive into Effective Prompt Engineering Techniques. *Appl. Sci.* **2025**, *15*, 1430. https://doi.org/10.3390/app15031430

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Recent advancements in Natural Language Processing (NLP) technology have been propelled at an unprecedented pace by the development of Large Language Models (LLMs).

These models are pre-trained on vast datasets, equipping them with the ability to recognize and comprehend complex language patterns. From question answering in conversational agents, text summarization, and language translation, to content generation and speech recognition, LLMs are integral to the progress of artificial intelligence, particularly in consumer electronics and technologies (CE/CT) [1]. In consumer applications, such as smart home assistants, voice-controlled devices, and interactive entertainment systems, LLMs contribute to more intuitive user experiences by enhancing natural language understanding and response generation.

The evolution of LLMs from simple statistical language models [2] to sophisticated transformer-based architectures like GPT-4 [3,4], PaLM [5], and LLaMA [6] marks a significant leap in their capabilities, influencing diverse CE/CT applications. Prominent models include OpenAI's GPT series [3,4], Meta's LLaMA family [7–9], Google's Gemini [2] and Gemma [10], as well as Mistral's recent advancements [11,12], all of which are continuously evolving to optimize their performance in consumer-facing technologies.

The capabilities of LLMs are grounded in Pre-trained Language Models (PLMs) and subsequent fine-tuning techniques. As PLMs have grown in scale, developing into highly capable LLMs, novel reasoning capabilities have emerged, further accelerating their utility in various applications. These abilities include In-Context Learning (ICL) [13], Instruction Following (IF) [14], and Step-by-Step Reasoning (SSR) [15], which are increasingly critical for enhancing consumer electronics interactions, such as improving response accuracy in smart assistants or optimizing content recommendations in entertainment systems.

Despite the advancements in PLMs, their accuracy often falls short, particularly when applied to specific datasets or tasks, and they are prone to generating incorrect answers that include non-existent information, a phenomenon known as hallucination. To address these issues, various Prompt Engineering (PE) techniques have recently been developed. Prompt Engineering plays a crucial role in determining the quality of the final output generated by an LLM by restructuring the prompt's structure and content to optimize the model's output. Notably, Prompt Engineering allows for the extraction of desired responses during the inference stage without the need for costly and time-consuming pre-training or fine-tuning stages, making it a highly advantageous technique for commercial applications [16].

One prominent approach within Prompt Engineering is the Retrieval-Augmented Generation (RAG) technique. RAG enhances the current data by combining a retrieval step, where external knowledge is leveraged to supplement the information, with a generation step that produces the final output. This integration improves the accuracy of information retrieval and enables the model to provide more accurate responses to queries that involve up-to-date information or rare facts.

In contrast, the Chain of Thought (CoT) technique is designed to guide language models to explicitly articulate intermediate steps or logical reasoning processes when solving complex problems. For instance, by structuring prompts to include the step-by-step solution process for a math problem, CoT enables the evaluation of both the correctness of the final response and the appropriateness of the reasoning that led to it.

The In-Context Learning (ICL) technique, as its name suggests, involves learning meaning within the context provided by the prompt. It enables the model to generate relevant responses by understanding the context in the prompt. This is typically achieved by providing examples, which guide the model in generating appropriate answers. Depending on the number of examples provided, the technique is categorized into zero-shot learning (no examples), one-shot learning (one example), and few-shot learning (multiple examples).

Additionally, the Step-by-Step Reasoning (SSR) technique guides the model to arrive at an answer through a series of intermediate steps when solving complex problems. This

approach can be particularly useful in tasks that require multi-step thinking, such as solving math problems, logical reasoning, or complex decision-making processes [15].

Lastly, the Tree of Thought (ToT) technique involves branching out multiple possible thought processes in a tree structure to explore different solutions to a complex problem. This method is especially effective in situations that require multi-stage reasoning or where multiple solutions are possible [17].

This paper investigates the optimal Prompt Engineering strategies that maximize user satisfaction, i.e., response accuracy, while minimizing resource requirements for real-world LLM service deployment. The study focuses on applying Prompt Engineering techniques without fine-tuning foundation models, enabling efficient service implementation. To achieve this, we examine the application strategies of widely used Prompt Engineering methods, including RAG, CoT, ICL, SSR, and ToT. These techniques were applied to leading LLM technologies, namely Llama3, Mistral, and Gemma2, to evaluate their performance. For performance evaluation, we utilized diverse metrics, including BLEU, ROUGE, METEOR, BLEURT, and BERTScore, to closely approximate user satisfaction. Benchmark datasets such as the AI2 Reasoning Challenge (ARC), HellaSwag, Massive Multitask Language Understanding (MMLU), TruthfulQA, Winogrande, and Grade School Math (GSM8K) were employed to analyze the natural language generation performance across a variety of topics. Based on these evaluations, we propose a PE (Prompt Engineering) strategy for optimizing performance across different datasets and LLMs.

2. Related Works

Prompt Engineering is a crucial technique for determining the quality of responses generated by LLMs. It optimizes model outputs by adjusting the structure and content of prompts during the inference stage, without requiring additional pre-training or fine-tuning of the model. Initially, Prompt Engineering primarily involved simple question-and-answer formats, but recent advancements have led to the development of more complex prompt structures designed to solve intricate problems [18]. Early research in this field, spearheaded by OpenAI with the development of the GPT-3 model, explored various Prompt Engineering methods. Among these, techniques for designing ICL-based few-shot learning prompts emerged as a way to obtain desired answers from LLMs without additional training [13]. The success of ICL has since led to the exploration of various derivatives, such as PICL [19], MetaICL [20], KATE [21], SG-ICL [22], and GlobalE&LocalE [23]. While early ICL research focused on optimizing ICL performance through prompt reconfiguration alone, the PICL [19] approach proposed collecting related contextual data and reconstructing corpora to pre-train the model, thereby enabling it to learn how to infer through prompt demonstrations in advance. In contrast, the MetaICL [20] approach introduced an additional phase between pre-training and ICL inference called continual training, also referred to as model warming. This optional procedure for ICL involves adjusting the LLM by modifying or adding to its parameters before inference. Many studies have also shown that ICL performance heavily depends on the configuration of demonstrations, including the selection, format, and order of the examples. KATE [21] investigated the selection of demonstration examples, SG-ICL [22] focused on the format, and GlobalE&LocalE [23] explored the order of demonstrations.

Research has also been conducted on techniques that encourage models to express intermediate steps or logical reasoning alongside responses to solve complex problems. Notable methods in this area include CoT, Self-Consistency, SSR, and ToT. The Self-Consistency technique optimizes the model to produce consistent and reliable outputs across varied inputs that convey the same meaning. The ToT approach involves branching out multiple reasoning steps required to solve a problem into various thought trees. Each branch

represents a different line of reasoning or assumption, allowing the exploration of diverse possibilities or hypotheses. The model then evaluates the relevance of these branches to the query, considering real-time factors to derive the most plausible result [17].

Recently, Prompt Engineering techniques have been developed to address the issue of hallucinations, a major challenge for LLMs where the model generates incorrect answers, including non-existent information [24]. Hallucination issues often arise from the lack of up-to-date knowledge in the training data or restricted access to specific cases or personal information. One of the most actively researched techniques to mitigate hallucinations is RAG [25]. RAG addresses this issue by extracting a query from the input prompt, using that query to retrieve information from external knowledge sources (e.g., search engines or knowledge graphs), and then augmenting the original prompt with this information or extracting relevant details provided in the prompt. The augmented prompt is then fed back into the LLM to generate the final response. RAG systems effectively resolve hallucinations through this process of retrieval, augmentation, and generation. The success of RAG has led to the development of related variants, including the LLM-Augmenter [26], which performs information retrieval before AI text generation; Knowledge Retrieval [27], which retrieves information during generation; RARR [28], which performs retrieval after generation; and the Chain-of-Verification (CoVe) technique [29], which incorporates self-refinement through feedback and logical reasoning.

The Prompt Engineering techniques described above are summarized in Table 1.

Table 1. Summary of Prompt Engineering techniques.

Techniques	Key Characteristics
ICL [13,19–23]	Focuses on few-shot learning by providing demonstrations within prompts. Includes derivatives such as PICL (contextual data collection and reconstruction), MetaICL (continual training phase), KATE (demonstration example selection), SG-ICL (format of demonstrations), and GlobalE&LocalE (demonstration order).
CoT [30,31]	Encourages intermediate reasoning steps for solving complex problems. Demonstrated to improve logical reasoning capabilities in LLMs.
Self-Consistency [31]	Optimizes the model to produce consistent and reliable outputs across varied inputs conveying the same meaning.
SSR [31]	Promotes systematic problem-solving by explicitly breaking down the reasoning process into individual steps.
ToT [17,31]	Represents reasoning as a tree structure with branches exploring diverse hypotheses or possibilities, aiding in logical exploration and plausibility evaluation.
RAG [24–29]	Addresses hallucination issues by retrieving external information (e.g., from knowledge graphs or search engines) and augmenting prompts with this information. Includes variants such as the LLM-Augmenter (retrieval before generation), Knowledge Retrieval (retrieval during generation), RARR (retrieval after generation), and CoVe (self-refinement via feedback).

Lastly, there have been studies addressing issues related to model alignment, demonstration formatting, and the theoretical underpinnings of prompting using the aforementioned Prompt Engineering techniques [30–32]. The study by [30] proposed efficient prompt construction methods such as Instruction+Question, Instruction+Input, and Question+Answer. It also identified the limitations of LLMs and suggested strategies to overcome these limitations using CoT-based Prompt Engineering. Similarly, ref. [31] explored effective prompt construction techniques and discussed optimization strategies for LLMs through approaches such as CoT, Self-Consistency, General Knowledge, Least-to-Most

Prompting, ToT, GoT, Decomposed Prompting, and Active Prompting. The study by [32] provided a SWOT analysis of LLM technologies and presented the mathematical theoretical underpinnings of various Prompt Engineering techniques. However, none of these studies have focused on deriving the optimal Prompt Engineering techniques based on performance analysis using diverse metrics tailored to the characteristics of LLMs and datasets.

The main contributions of this paper are as follows:

- A technical analysis of various LLMs, Prompt Engineering techniques for performance enhancement, related datasets, and evaluation metrics.
- A comparative performance analysis of key Prompt Engineering techniques across different LLMs and datasets, using various evaluation metrics.
- Derivation of performance optimization strategies based on the comparative analysis, tailored to specific application domains.

In Section 3, we describe the LLMs utilized in this study. Section 4 provides an overview of the benchmark datasets employed, while Section 3.2 explains the performance metrics used for evaluation. In Section 5, we present the experimental results based on these parameters and propose a PE Strategy for optimizing LLM performance. Finally, Section 6 offers the conclusion.

3. System Models and Evaluation Metrics

3.1. System Models

This paper conducts experiments on LLMs, specifically focusing on Gemma2, Llama3, and Mistral—three of the most high-performance, widely used, and recently released technologies.

3.1.1. Gemma2

Gemma2, released by Google on 27 June 2024, is the latest open large language model, available in two sizes, i.e., 9 billion and 27 billion parameters, based on the Hugging Face platform. Each model includes pre-training, fine-tuning, and alignment phases. As a lightweight version of the larger Gemini model, Gemma2 offers significant enhancements. The key features of the Gemma2 model are summarized in Table 2.

Table 2. Summary of the model architecture, pre-training, and post-training techniques of major LLMs.

	Gemma2 (9B/27B)	**Llama3 (8B/70B)**	**Mistral (7B)**
Model Architecture	Decoder-only Transformer Rotary Position Embeddings (RoPE) Approx. GeGLU non-linearity Local Sliding Windows Global Attention Logit Soft-Capping Grouped-Query Attention (GQA)	Decoder-only Transformer Rotary Position Embeddings (RoPE) SwiGLU Grouped-Query Attention (GQA)	Decoder-only Transformer Grouped-Query Attention Sliding Window Attention Rolling Buffer Cache Pre-fill and Chunking
Pre-Training	Curation and filtering of training corpus Vocab Size: 256 K Knowledge Distillation	Curation and filtering of training corpus Vocab Size: 128 K	Curation and filtering of training corpus Vocab Size: 32 K
Post-Training	Supervised Fine-Tuning (SFT) RL with Human Feedback (RLHF) Model Merging	Supervised Fine-Tuning (SFT) RL with Human Feedback (RLHF) Rejection Sampling (RS) Direct Preference Optimization (DPO)	Supervised Fine-Tuning (SFT) System Prompt to enforce guardrails

The model architecture is based on a Transformer with a Decoder-Only structure, utilizing Grouped Query Attention (GQA) with 8 key-valued heads and 16 heads. Rotary Position Embeddings (RoPE) [33] and approximated GeGLU non-linearity [34] are employed. Additionally, Local Sliding Windows and Global Attention are alternated in every

layer, and logits in each attention layer and the final layer are constrained to remain within a soft-cap range.

The training methodology is divided into the following two stages: Pre-Training and Post-Training. In the Pre-Training stage, data filtering is first applied to address privacy and safety issues, and the filtered data is then converted into discrete tokens from a large multilingual text corpus. Instead of next-token prediction, the model is trained using Knowledge Distillation based on the probability of each token as provided by the teacher model.

During the Post-Training stage, the pre-trained models are fine-tuned into Supervised Fine-Tuned models. Subsequently, Reinforcement Learning with Human Feedback (RLHF) is applied to these models, where the reward model is trained on labeled English preference data and the policy is based on the same prompts used in the supervised fine-tuning (SFT) phase. Finally, the models obtained after each phase are averaged to enhance overall performance.

3.1.2. Llama3

Llama3 is the latest large language model developed by Meta, released on 18 April 2024. Building on the foundation of its predecessor, Llama2, this model is available in two sizes. i.e., 8 billion and 70 billion parameters [9]. The key features of the Llama3 model are summarized in Table 2.

The model architecture is based on a Transformer with a Decoder-Only structure, utilizing GQA with 8 key-valued heads. During the Pre-Training stage, a large multilingual text corpus is first converted into discrete tokens, and then a LLM is pre-trained on the resulting data to perform next-token prediction.

While the pre-trained language model has a rich understanding of language, it does not yet fully follow instructions or generate clear responses. In the Post-Training stage, the model is aligned with human feedback over multiple rounds, each involving SFT on instruction-tuning data and Direct Preference Optimization (DPO). Additionally, specialized training is introduced for specific domains such as coding and reasoning.

3.1.3. Mistral

Mistral is a high-performance language model developed by the French AI startup Mistral AI. The Mistral model is a language model with 7 billion parameters, leveraging cutting-edge technologies such as GQA and Sliding Window Attention (SWA) to enhance efficiency [11,12]. The key technical features are summarized in Table 2.

The model architecture is based on a Transformer with a Decoder-Only structure, incorporating GQA with 8 key-valued heads and 32 heads, as well as a SWA mechanism with a size of 4096 to accelerate inference speed. Additionally, to optimize cache usage, techniques such as Rolling Buffer Cache, Pre-fill, and Chunking are employed.

During the Pre-Training stage, a large multilingual text corpus is first converted into discrete tokens, and then the resulting data is used to pre-train a LLM for next-token prediction.

In the Post-Training stage, Instruction Fine-tuning is conducted through SFT using Hugging Face's Instruction dataset. Finally, the model performs fine-grained content moderation by leveraging system prompts to optionally enforce constraints on its output.

3.2. Performance Metric

This paper selects and utilizes some of the most common and widely used metrics for performance evaluation, including BLEU (Bilingual Evaluation Understudy), ROUGE (Recall-Oriented Understudy for Gisting Evaluation), METEOR (Metric for Evaluation of Translation with Explicit Ordering), BLEURT (Bilingual Evaluation Understudy with

Representations from Transformers), and BERT (Bidirectional Encoder Representations from Transformers) Score, in order to conduct a comprehensive analysis from various performance evaluation perspectives.

3.2.1. BLEU

The BLEU score is a metric developed to evaluate the quality of machine translation by measuring the n-gram overlap between the translated sentence and reference sentences. The BLEU score is calculated using the following formula:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \tag{1}$$

where p_n represents the n-gram precision, w_n denotes the weight, and BP refers to the brevity penalty. BLEU is primarily used to assess the fluency and accuracy of translated text, where a higher score indicates a closer match between the translated text and the reference text [35].

3.2.2. ROUGE

The ROUGE score is primarily used to measure the similarity between generated text and reference text in summarization tasks. There are several variants of ROUGE, including ROUGE-N, ROUGE-L, and ROUGE-W. ROUGE-N is based on n-gram overlap, while ROUGE-L relies on the Longest Common Subsequence (LCS). ROUGE-W is a weighted version of ROUGE-L, where greater weights are assigned to longer common subsequences. ROUGE is useful for evaluating the coverage and consistency of generated summaries and is frequently employed in multi-document summarization evaluations [36]. In this study, for the short-answer datasets, we based our performance measurement on ROUGE-N with $N = 1$ (unigrams). The formula for ROUGE-N is as follows:

$$\text{ROUGE-N} = \frac{\sum_{s \in \{\text{RS}\}} \sum_{\text{gram}_n \in s} \text{Count}_m(\text{gram}_n)}{\sum_{s \in \{\text{RS}\}} \sum_{\text{gram}_n \in s} \text{Count}(\text{gram}_n)} \tag{2}$$

where gram_n is n-grams, RS is the Reference Summaries, $\text{Count}_\text{m}(\text{gram}_n)$ is the number of n-grams in the generated text that match the n-grams in the reference text, and $\text{Count}(\text{gram}_n)$ is the total number of n-grams in the reference text.

3.2.3. METEOR

The METEOR metric, unlike BLEU, accounts for factors beyond simple n-gram matching, including synonymy, stemming, and word order alignment. METEOR employs a harmonic mean of unigram precision and recall, placing a higher weight on recall in its evaluation. By incorporating considerations such as synonyms and morphological analysis, METEOR enhances the flexibility of word matching and is known to correlate more closely with human judgments compared to BLEU. This allows for a more nuanced assessment of translation quality [37].

$$\begin{aligned} \text{METEOR} &= F_{\text{mean}} \times (1 - P_{\text{penalty}}), \\ F_{mean} &= \frac{10 \cdot P \cdot R}{R + 9 \cdot P}, \\ P_{penalty} &= 0.5 \cdot \left(\frac{\text{number of chunks}}{\text{number of aligned words}}\right)^3 \end{aligned} \tag{3}$$

where P is the Precision, R is the Recall, and $P_{penalty}$ is the fragmentation penalty for penalizing word order errors. These metrics are calculated using following equations:

$$P = \frac{\text{Number of aligned words}}{\text{Number of words in the candidate translation}}, \tag{4}$$

$$R = \frac{\text{Number of aligned words}}{\text{number of words in the reference translation}}. \tag{5}$$

3.2.4. BLEURT

Unlike traditional metrics such as BLEU, which rely on n-gram overlap, BLEURT is a metric designed to account for semantic similarity, contextual understanding, and fluency. This results in an evaluation metric that more closely aligns with human assessments [38]. Specifically, BLEURT leverages BERT (Bidirectional Encoder Representations from Transformers) embeddings and is fine-tuned for specific text evaluation tasks. The model is trained to predict a score that reflects how well the generated text aligns with the reference text, based on the following equation:

$$\text{BLEURT}(c, r) = f(\text{BERT}(c), \text{BERT}(r), \text{BERT}(ctxt)),$$

where c is the generated text, r is the reference text, and $ctxt$ is the context which may include additional information or surrounding text that can influence the score. $\text{BERT}(x)$ represents the BERT embedding of the text x and $f(\cdot)$ is a function that combines these embeddings and outputs a score reflecting the similarity between c and r.

3.2.5. BERT Score

BERTScore is an evaluation metric designed to measure semantic similarity between texts by utilizing the BERT model to generate embeddings for each word, which are then used to compute the semantic similarity between the reference and generated sentences [39]. BERTScore is based on the cosine similarity between the embedding vectors of each word, forming word pairs between the reference and generated sentences to calculate the similarity. Mathematically, it can be expressed as follows:

$$\begin{aligned} P &= \frac{1}{|X|} \sum_{X} \max_{Y} \text{cosine}(X, Y), \\ R &= \frac{1}{|Y|} \sum_{Y} \max_{X} \text{cosine}(X, Y), \\ \text{BERTScore} &= 2 \times \frac{P \cdot R}{P + R}, \end{aligned} \tag{6}$$

where P is the precision, R is the recall, X is the embedding vector of the generated text x, Y is the embedding vector of the reference text y, and $\text{cosine}(X, Y)$ represents the cosine similarity between the embeddings of the tokens from the generated and reference texts.

4. Dataset

In this study, we selected benchmark datasets from three categories to comprehensively evaluate various language capabilities. For the general natural language understanding category, we chose the MMLU and TruthfulQA datasets. For the reasoning ability category, we selected the ARC, HellaSwag and Winogrande datasets. For the mathematics problem-solving category, we used the GSM8K dataset. These categories were carefully chosen to ensure a balanced assessment across general natural language understanding, reasoning ability, and mathematics problem-solving domains.

4.1. ARC

The ARC dataset consists of 11,119 training, 2992 validation, and 4541 test examples, and is composed of multiple-choice questions with four options each, covering science topics typically taught at the elementary and middle school levels in the United States, such as biology, physics, chemistry, and earth science. The dataset is divided into two categories as follows: ARC-Easy, which includes questions that require basic information retrieval, and ARC-Challenge, which contains more difficult questions that require complex reasoning and inference. This dataset was designed to evaluate the extent to which AI models can perform complex reasoning and understanding beyond simple information retrieval [40].

4.2. HellaSwag

The HellaSwag dataset consists of 39,905 training, 5000 validation, and 10,042 test examples. It is designed to evaluate AI models' natural language understanding by testing their ability in commonsense reasoning and contextual understanding through sentence completion and multiple-choice questions. This dataset includes contexts from various domains, such as Wikipedia and instructional videos, and requires models to predict the most appropriate sentence continuation among the provided choices for each context [41].

4.3. MMLU

The MMLU dataset consists of 57,243 training, 10,000 validation, and 15,000 test examples. Developed as part of Facebook AI's efforts to advance human language understanding and generation capabilities, the dataset includes multiple-choice questions across a wide range of subjects, including science, humanities, and professional fields. It is designed to assess how broadly and deeply AI models can understand and apply knowledge across various domains [42].

4.4. TruthfulQA

The TruthfulQA dataset consists of 8170 training, 1024 validation, and 2035 test examples. It is designed to evaluate the ability of language models to generate factual and truthful responses. The dataset includes questions that often use misleading or ambiguous phrasing, or that require the model to acknowledge its lack of knowledge rather than fabricating an answer. This dataset is used to assess the limitations of a model's truthfulness and its ability to handle subtlety in information retrieval and response generation [43].

4.5. Winogrande

The Winogrande dataset consists of 40,398 training, 1267 validation, and 3545 test examples and is designed to train and test the commonsense reasoning abilities within AI systems. Inspired by the Winograd Schema Challenge, it was developed with a larger scale and a more diverse set of items to more thoroughly evaluate model performance. The dataset comprises sentences with blanks that must be filled by selecting the correct word from two given options, testing the model's ability to perform contextual and commonsense reasoning [44].

4.6. GSM8K

The GSM8k dataset consists of 7432 training, 800 validation, and 1319 test examples, and is designed to test the mathematical reasoning abilities of AI systems. This dataset is a collection of elementary-level math problems covering fundamental concepts such as arithmetic, algebra, geometry, and word problems. It is used to evaluate AI models not only on their ability to compute answers but also on their performance in understanding the text context of problems, which often involves natural language comprehension and multi-step reasoning [45].

5. Simulation Result

Table 3 presents the experimental setup used in this study. The experiments were conducted using Llama3-8B, Gemma2-9B, and Mistral-7B as the LLMs, with a focus on analyzing the performance of ICL(1 Shot), CoT, RAG(1 Shot), ToT, SSR, and their combined configurations. Sections 5.1–5.6 detail the performance evaluation and comparative analysis across six datasets, based on the experimental environment outlined in Table 3. The performance results were generated based on the prompt formats created according to the respective Prompt Engineering techniques, with parameters set as Temperature = 0.1 and Top-P = 0.9. Additionally, the applied Prompt Engineering techniques are denoted by the following abbreviations and may be used interchangeably:

B: Base, I: ICL, C: CoT, S: SSR, T: ToT,

R(B): Basic RAG, R(C): CoT based RAG

For performance analysis, we conducted prompt fine-tuning on pre-trained models—LLama3 8B, Mistral 7B, and Gemma2 9B—utilizing the computing resources of the NVIDIA H100 Tensor Core GPU (Santa Clara, CA, USA). Finally, in Section 5.7, we analyzed the Prompt Engineering techniques that yielded the best performance for each dataset across the different LLMs, leading to the derivation of Prompt Engineering strategies aimed at optimizing performance.

Table 3. Experimental setup.

Model					
Llama3 8B		Gemma2 9B		Mistral 7B	
Prompt Engineering					
B: Base	**I:** ICL	**C:** CoT	**R:** Rag	**T:** ToT	**S:** SSR
Metric					
BLUE	ROUGE	METEOR	BLUERT	BERT	-
Dataset					
ARC	HellaSwag	MMLU	TruthfulQA	Winogrande	GSM8K

5.1. ARC

Tables 4 and 5 present the performance of Llama3, Gemma2, Mistral, and various Prompt Engineering combinations on the ARC dataset, measured using the BLEU, ROUGE, METEOR, BLEURT, and BERTScore metrics. The bolded values in the tables indicate the highest performance achieved for each evaluation metric within each LLM model. Furthermore, the bold formatting convention is consistently applied to the performance result tables of other datasets as well. As shown in these tables, the ARC dataset, which consists of multiple-choice questions and answers on scientific topics, achieves the best performance with combinations of SSR techniques and CoT-based methods such as R(B), R(C), C, and T.

Moreover, it is evident that the optimal combination of Prompt Engineering techniques varies depending on the LLM and the evaluation metric. For Llama3, using CoT alone generally yields the best results. In contrast, for Gemma2, the combination of SSR, CoT, and ICL demonstrates superior performance, while for Mistral, the combination of SSR, ToT, CoT, and ICL achieves the best outcomes.

Fundamentally, as observed in Tables 4 and 5, Gemma2 consistently outperforms other models across all five evaluation metrics when no Prompt Engineering techniques

are applied. Even when considering all possible Prompt Engineering techniques, Gemma2 remains the top-performing LLM, with a significant performance gap compared to the second-best model, Llama3. Additionally, Llama3 consistently outperforms Mistral across all five evaluation metrics.

Table 4. Performances of BLUE, ROUGE, and METEOR according to LLMs and Prompt Engineering techniques on the ARC dataset.

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEU				ROUGE				METEOR		
B	2.2864	4.4464	1.4115	**B**	0.1955	0.2439	0.1426	**B**	0.1514	0.2007	0.1375
I	2.4251	5.5576	1.5623	**I**	0.1750	0.2575	0.0990	**I**	0.1237	0.2156	0.0971
C	0.0000	0.0000	1.8549	**C**	**0.6553**	0.8340	0.4298	**C**	**0.3277**	0.4170	0.2163
C, I	0.0000	0.0000	0.0000	**C, I**	0.3787	0.8298	0.2681	**C, I**	0.1872	0.4149	0.1340
R(B)	2.2759	4.8229	1.4879	**R(B)**	0.2073	0.2561	0.1009	**R(B)**	0.1501	0.2094	0.0929
R(C)	0.0000	0.0000	1.5619	**R(C)**	0.5830	**0.8383**	0.3438	**R(C)**	0.2915	**0.4191**	0.1700
T, B	2.0589	3.1321	1.2867	**T, B**	0.2002	0.2345	0.1499	**T, B**	0.1499	0.1843	0.1552
T, I	2.1093	4.3401	1.0184	**T, I**	0.1978	0.2487	0.1432	**T, I**	0.1433	0.1997	0.1368
T, C	0.0000	0.0000	3.4968	**T, C**	0.6511	0.8383	0.4199	**T, C**	0.3255	**0.4191**	0.2104
T, C, I	0.0000	0.0000	2.3658	**T, C, I**	0.5362	0.8298	0.2298	**T, C, I**	0.2681	0.4149	0.1149
T, R(B)	2.0709	4.7891	1.5037	**T, R(B)**	0.2023	0.2434	0.1407	**T, R(B)**	0.1462	0.2000	0.1322
T, R(C)	0.0000	0.0000	3.9588	**T, R(C)**	0.5532	0.8340	0.3574	**T, R(C)**	0.2766	0.4170	0.1785
S, B	2.1450	3.0319	0.6572	**S, B**	0.1927	0.2346	0.1290	**S, B**	0.1499	0.1843	0.1313
S, I	1.2337	4.5014	0.4917	**S, I**	0.1399	0.2444	0.0661	**S, I**	0.0866	0.1962	0.0716
S, C	0.0000	**21.4646**	1.2169	**S, C**	0.6170	0.8255	0.3887	**S, C**	0.3085	0.4128	0.1954
S, C, I	0.0000	0.0000	1.6279	**S, C, I**	0.5702	0.8340	0.2979	**S, C, I**	0.2851	0.4170	0.1489
S, R(B)	1.2236	4.2853	1.4485	**S, R(B)**	0.1714	0.2500	0.1352	**S, R(B)**	0.1197	0.2007	0.1247
S, R(C)	**19.5521**	21.3812	0.8924	**S, R(C)**	0.5660	0.8128	0.3234	**S, R(C)**	0.2830	0.4064	0.1591
S, T, B	2.3439	2.7917	0.9111	**S, T, B**	0.1887	0.2355	0.1349	**S, T, B**	0.1433	0.1816	0.1352
S, T, I	0.7283	3.6081	0.1601	**S, T, I**	0.1157	0.2400	0.0818	**S, T, I**	0.0869	0.1841	0.0905
S, T, C	0.0000	0.0000	1.7047	**S, T, C**	0.6426	0.8340	0.2468	**S, T, C**	0.3213	0.4170	0.1234
S, T, C, I	0.0000	0.0000	**16.4416**	**S, T, C, I**	0.5489	0.8170	**0.4340**	**S, T, C, I**	0.2745	0.4085	**0.2170**
S, T, R(B)	1.0221	4.4266	0.6973	**S, T, R(B)**	0.1764	0.2460	0.1473	**S, T, R(B)**	0.1285	0.1976	0.1365
S, T, R(C)	0.0000	0.0000	1.5277	**S, T, R(C)**	0.5447	0.8255	0.3263	**S, T, R(C)**	0.2723	0.4128	0.1612

Table 5. Performances of BLUERT and BERTScore according to LLMs and Prompt Engineering techniques on the ARC dataset.

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEURT				BERTScore		
B	0.3167	0.3587	0.2857	**B**	0.5969	0.6277	0.5364
I	0.2760	0.3725	0.2188	**I**	0.5756	0.6327	0.4701
C	**0.4662**	0.5336	0.3678	**C**	0.9690	0.9811	0.9019
C, I	0.3450	0.5343	0.3050	**C, I**	0.9363	0.9844	0.9193
R(B)	0.3107	0.3663	0.2238	**R(B)**	0.6038	0.6455	0.4786
R(C)	0.4427	0.5378	0.3310	**R(C)**	0.9651	0.9848	0.9047
T, B	0.3172	0.3454	0.3002	**T, B**	0.5971	0.6276	0.5282
T, I	0.3221	0.3668	0.2825	**T, I**	0.5955	0.6327	0.4846
T, C	0.4654	0.5367	0.3706	**T, C**	**0.9704**	0.9822	0.9296

Table 5. *Cont.*

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEURT				BERTScore		
T, C, I	0.4132	0.5366	0.2850	**T, C, I**	0.9600	0.9858	0.9146
T, R(B)	0.3050	0.3543	0.2750	**T, R(B)**	0.6052	0.6350	0.5223
T, R(C)	0.4269	0.5365	0.3435	**T, R(C)**	0.9624	0.9846	0.9213
S, B	0.3133	0.3507	0.2780	**S, B**	0.5842	0.6189	0.4968
S, I	0.2380	0.3603	0.1944	**S, I**	0.5481	0.6172	0.3805
S, C	0.4526	0.5303	0.3499	**S, C**	0.9666	0.9788	0.8947
S, C, I	0.4257	**0.5379**	0.3285	**S, C, I**	0.9607	**0.9859**	0.9060
S, R(B)	0.2770	0.3581	0.2639	**S, R(B)**	0.5819	0.6385	0.4966
S, R(C)	0.4332	0.5292	0.3335	**S, R(C)**	0.9613	0.9814	0.8764
S, T, B	0.3144	0.3444	0.2792	**S, T, B**	0.5883	0.6244	0.4907
S, T, I	0.2110	0.3537	0.2250	**S, T, I**	0.4742	0.6204	0.3728
S, T, C	0.4615	0.5341	0.2906	**S, T, C**	0.9699	0.9812	0.9121
S, T, C, I	0.4284	0.5306	**0.3977**	**S, T, C, I**	0.9626	0.9845	**0.9413**
S, T, R(B)	0.2873	0.3524	0.2821	**S, T, R(B)**	0.5854	0.6347	0.4965
S, T, R(C)	0.4280	0.5334	0.3438	**S, T, R(C)**	0.9618	0.9836	0.9084

5.2. GSM8K

Tables 6 and 7 present the performance of Llama3, Gemma2, Mistral, and various Prompt Engineering combinations on the GSM8K dataset, measured using the BLEU, ROUGE, METEOR, BLEURT, and BERTScore metrics.

As shown in these tables, the GSM8K dataset, which involves elementary-level math problems, described in natural language and requiring multi-step reasoning, achieves the best performance with combinations of Step-by-Step Reasoning techniques, CoT-based methods such as R(B) and R(C), CoT techniques, ToT, and ICL.

The optimal combinations of these Prompt Engineering techniques vary depending on the LLM and the evaluation metric. For Llama3, although the optimal Prompt Engineering technique varies across different evaluation metrics, combinations of SSR and CoT-based RAG generally yield the best performance. For Gemma2, combinations of CoT and ICL tend to provide the best results in most cases, while for Mistral, combinations of ToT, CoT, and ICL consistently deliver the best performance.

Fundamentally, as seen in Tables 6 and 7, when no Prompt Engineering techniques are applied, Llama3 outperforms others in three out of five evaluation metrics, while Gemma2 performs best in two metrics. Even when considering all possible Prompt Engineering techniques, Llama3 leads in three evaluation metrics, while Gemma2 leads in two. This indicates that Llama3 generally delivers the best performance on the GSM8k dataset. Finally, Mistral ranks third, showing a significant performance gap compared to Llama3 and Gemma2.

Table 6. Performances of BLUE, ROUGE, METEOR according to LLMs and Prompt Engineering techniques on the GSM8k dataset.

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEU				ROUGE				METEOR		
B	10.1785	8.2318	6.2994	**B**	0.4283	0.4434	0.3382	**B**	0.2512	0.2379	0.1986
I	11.3725	10.3809	6.3203	**I**	0.4505	0.4471	0.2904	**I**	0.2680	0.2557	0.1799
C	11.2824	9.7220	6.6639	**C**	0.4401	0.4493	0.3157	**C**	0.2653	0.2527	0.1924

Table 6. *Cont.*

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEU				ROUGE				METEOR		
C, I	11.1692	11.6382	7.8677	**C, I**	0.4455	**0.4818**	0.3349	**C, I**	0.2637	**0.2784**	0.2060
R(B)	10.4893	10.0561	7.2215	**R(B)**	0.4304	0.4513	0.3164	**R(B)**	0.2577	0.2514	0.1966
R(C)	12.0336	10.5243	7.4787	**R(C)**	0.4544	0.4537	0.3192	**R(C)**	0.2730	0.2576	0.1979
T, B	11.3938	8.8232	6.8033	**T, B**	0.4234	0.4550	0.3214	**T, B**	0.2654	0.2451	0.1948
T, I	12.1602	10.1051	6.1928	**T, I**	0.4411	0.4406	0.3072	**T, I**	0.2773	0.2517	0.1956
T, C	12.4078	10.2231	6.9790	**T, C**	0.4364	0.4609	0.3405	**T, C**	0.2755	0.2599	0.2091
T, C, I	12.4400	11.3522	**9.7138**	**T, C, I**	0.4511	0.4694	**0.3492**	**T, C, I**	0.2813	0.2738	**0.2345**
T, R(B)	12.2981	10.2974	7.5567	**T, R(B)**	0.4414	0.4540	0.3301	**T, R(B)**	0.2797	0.2595	0.2091
T, R(C)	13.6000	**11.6573**	9.4806	**T, R(C)**	0.4675	0.4718	0.3248	**T, R(C)**	**0.2919**	0.2738	0.2188
S, B	11.0489	7.3710	5.8964	**S, B**	0.4423	0.4427	0.3089	**S, B**	0.2683	0.2330	0.1859
S, I	11.2372	8.8757	4.0492	**S, I**	0.4309	0.4582	0.2516	**S, I**	0.2608	0.2540	0.1559
S, C	11.2324	9.6305	6.2983	**S, C**	0.4464	0.4522	0.3260	**S, C**	0.2738	0.2504	0.1969
S, C, I	12.2068	8.9984	6.1892	**S, C, I**	0.4503	0.4550	0.2950	**S, C, I**	0.2764	0.2499	0.1848
S, R(B)	11.4328	9.1563	6.5322	**S, R(B)**	0.4517	0.4593	0.3064	**S, R(B)**	0.2751	0.2546	0.1903
S, R(C)	12.4954	9.9989	8.0168	**S, R(C)**	**0.4717**	0.4628	0.3188	**S, R(C)**	0.2873	0.2551	0.2056
S, T, B	11.5911	7.8427	7.1353	**S, T, B**	0.4364	0.4566	0.3299	**S, T, B**	0.2762	0.2431	0.2019
S, T, I	12.9408	9.5542	5.7495	**S, T, I**	0.4522	0.4376	0.2702	**S, T, I**	0.2865	0.2474	0.1757
S, T, C	12.0767	9.5349	7.5402	**S, T, C**	0.4388	0.4570	0.3295	**S, T, C**	0.2794	0.2550	0.2044
S, T, C, I	12.7459	10.8192	8.4431	**S, T, C, I**	0.4421	0.4668	0.3271	**S, T, C, I**	0.2833	0.2662	0.2102
S, T, R(B)	11.6348	10.0010	7.8086	**S, T, R(B)**	0.4408	0.4515	0.3239	**S, T, R(B)**	0.2790	0.2568	0.2126
S, T, R(C)	**13.7994**	10.8942	9.0781	**S, T, R(C)**	0.4567	0.4650	0.3427	**S, T, R(C)**	0.2895	0.2664	0.2214

Table 7. Performances of BLUERT, BERTScore according to LLMs and Prompt Engineering techniques on the GSM8k dataset.

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEURT				BERTScore		
B	0.4063	0.4008	0.3691	**B**	0.8119	0.8515	0.7033
I	0.4176	0.3988	0.3568	**I**	**0.8662**	0.8603	0.6001
C	0.4148	0.3994	0.3756	**C**	0.8374	0.8648	0.6778
C, I	0.4154	**0.4154**	0.3741	**C, I**	0.8418	**0.8747**	0.6793
R(B)	0.4138	0.3994	0.3712	**R(B)**	0.8192	0.8609	0.6610
R(C)	0.4198	0.4075	0.3710	**R(C)**	0.8228	0.8460	0.6278
T, B	0.4074	0.4022	0.3751	**T, B**	0.8307	0.8664	0.6785
T, I	0.4146	0.3962	0.3675	**T, I**	0.8454	0.8438	0.6682
T, C	0.4155	0.4049	0.3816	**T, C**	0.8436	0.8711	0.7051
T, C, I	0.4199	0.4035	**0.3873**	**T, C, I**	0.8594	0.8716	0.6974
T, R(B)	0.4149	0.3989	0.3771	**T, R(B)**	0.8613	0.8677	**0.7077**
T, R(C)	0.4245	0.4100	0.3778	**T, R(C)**	0.8577	0.8702	0.6618
S, B	0.4168	0.3977	0.3727	**S, B**	0.8487	0.8416	0.6495
S, I	0.4135	0.4073	0.3503	**S, I**	0.8429	0.8592	0.5127
S, C	0.4200	0.4005	0.3800	**S, C**	0.8582	0.8613	0.6733
S, C, I	0.4216	0.4052	0.3584	**S, C, I**	0.8538	0.8531	0.6410
S, R(B)	0.4195	0.4067	0.3689	**S, R(B)**	0.8597	0.8536	0.6728

Table 7. *Cont.*

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEURT				BERTScore		
S, R(C)	**0.4251**	0.4100	0.3765	**S, R(C)**	0.8572	0.8542	0.6610
S, T, B	0.4162	0.4052	0.3813	**S, T, B**	0.8540	0.8598	0.6745
S, T, I	0.4195	0.3948	0.3600	**S, T, I**	0.8611	0.8182	0.6019
S, T, C	0.4176	0.4051	0.3853	**S, T, C**	0.8429	0.8615	0.6689
S, T, C, I	0.4192	0.4053	0.3805	**S, T, C, I**	0.8514	0.8676	0.6701
S, T, R(B)	0.4176	0.3999	0.3781	**S, T, R(B)**	0.8634	0.8470	0.7072
S, T, R(C)	0.4240	0.4091	0.3812	**S, T, R(C)**	0.8597	0.8564	0.7068

5.3. HellaSwag

Tables 8 and 9 present the performance of Llama3, Gemma2, Mistral, and various combinations of Prompt Engineering techniques on the HellaSwag dataset, as measured by the BLEU, ROUGE, METEOR, BLEURT, and BERTScore metrics. As seen in these tables, the HellaSwag dataset, which evaluates natural language understanding, common sense reasoning, and situational comprehension, shows that combinations of CoT, ICL, ToT, and CoT-based RAG techniques yield the best performance.

Moreover, the optimal combination of these Prompt Engineering techniques varies depending on the LLM and the evaluation metric, while the best-performing Prompt Engineering approach changes across metrics for all models as follows: Llama3, Gemma2, and Mistral. Llama3 generally achieves the best performance with standalone CoT or combinations of CoT with ICL, ToT, and SSR techniques. For Gemma2, the combination of CoT-based RAG and ToT techniques yields the best results, while Mistral generally performs best with combinations of CoT and ICL.

Fundamentally, as observed in Tables 8 and 9, when no Prompt Engineering techniques are applied, Llama3 typically delivers the best performance, leading in four out of the five evaluation metrics. Gemma2 only slightly outperforms Llama3 in the BLEURT metric.

However, when all possible Prompt Engineering techniques are considered, Gemma2 emerges as the best-performing LLM, leading in all five evaluation metrics with a substantial performance gap over Llama3, the second-best model. Notably, although Gemma2's base PLM initially showed the lowest performance in BLEU, ROUGE, METEOR, and BERTScore, its performance was significantly enhanced through Prompt Engineering strategies, ultimately making it the top-performing LLM. This demonstrates that appropriate combinations of Prompt Engineering techniques can significantly improve the performance of even a baseline PLM.

Table 8. Performances of BLUE, ROUGE, METEOR according to LLMs and Prompt Engineering techniques on the HellaSwag dataset.

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEU				ROUGE				METEOR		
B	0.9938	0.4228	0.8254	**B**	0.1477	0.1335	0.1463	**B**	0.0951	0.0926	0.1034
I	0.1473	0.3997	0.2550	**I**	0.0855	0.1429	0.0942	**I**	0.0469	0.1047	0.0659
C	0.0193	0.0000	0.0531	**C**	0.3168	0.6521	0.2600	**C**	**0.1592**	0.3265	0.1312
C, I	**1.8112**	0.0000	0.0000	**C, I**	0.3604	0.7168	**0.3206**	**C, I**	0.1807	0.3586	0.1605
R(B)	0.8526	0.7209	0.6942	**R(B)**	0.1309	0.1488	0.1205	**R(B)**	0.0822	0.0999	0.0858
R(C)	1.0346	0.0000	**2.0251**	**R(C)**	0.3440	0.7158	0.2680	**R(C)**	0.1715	**0.3584**	0.1344
T, B	0.8721	0.2981	0.9997	**T, B**	0.1434	0.1290	0.1636	**T, B**	0.0999	0.0897	0.1222
T, I	1.0498	0.2970	0.7208	**T, I**	0.1399	0.1213	0.1335	**T, I**	0.1056	0.0854	0.0996

Table 8. *Cont.*

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEU				ROUGE				METEOR		
T, C	0.7567	**2.1914**	0.0888	**T, C**	**0.4604**	0.6989	0.3061	**T, C**	0.2305	0.3497	0.1535
T, C, I	0.0000	0.0000	0.0000	**T, C, I**	0.2633	0.7193	0.2937	**T, C, I**	0.1319	0.3599	0.1468
T, R(B)	0.7593	0.4324	1.1741	**T, R(B)**	0.1382	0.1373	0.1571	**T, R(B)**	0.0885	0.0917	0.1157
T, R(C)	0.0000	0.0000	0.7358	**T, R(C)**	0.3892	**0.7382**	0.2429	**T, R(C)**	0.1944	0.3693	0.1215
S, B	1.0413	0.4181	0.4664	**S, B**	0.1522	0.1345	0.1035	**S, B**	0.0982	0.0911	0.0705
S, I	0.0003	0.5170	0.2350	**S, I**	0.0204	0.1202	0.1059	**S, I**	0.0135	0.0842	0.0823
S, C	0.0054	1.5551	0.0240	**S, C**	0.1226	0.6381	0.3175	**S, C**	0.0641	0.3193	**0.1614**
S, C, I	0.0000	0.0000	0.5114	**S, C, I**	0.3116	0.6565	0.2992	**S, C, I**	0.1563	0.3283	0.1503
S, R(B)	0.7829	0.6365	1.3147	**S, R(B)**	0.1305	0.1395	0.1745	**S, R(B)**	0.0826	0.0952	0.1327
S, R(C)	0.5530	0.0000	0.9046	**S, R(C)**	0.3041	0.6775	0.3204	**S, R(C)**	0.1518	0.3387	0.1604
S, T, B	0.9205	0.3513	1.2806	**S, T, B**	0.1500	0.1355	0.1666	**S, T, B**	0.0979	0.0919	0.1271
S, T, I	1.1919	0.5674	1.1760	**S, T, I**	0.1496	0.1388	0.1510	**S, T, I**	0.1091	0.0971	0.1250
S, T, C	0.7736	0.9067	0.0124	**S, T, C**	0.4251	0.6585	0.2467	**S, T, C**	0.2125	0.3295	0.1252
S, T, C, I	0.0000	0.0000	0.0898	**S, T, C, I**	0.3718	0.7222	0.2683	**S, T, C, I**	0.1862	0.3614	0.1346
S, T, R(B)	0.9430	0.5889	1.3780	**S, T, R(B)**	0.1513	0.1473	0.1792	**S, T, R(B)**	0.0981	0.0993	0.1415
S, T, R(C)	0.0000	0.0000	0.3419	**S, T, R(C)**	0.3454	0.7133	0.2687	**S, T, R(C)**	0.1727	0.3569	0.1348

Table 9. Performances of BLUERT, BERTScore according to LLMs and Prompt Engineering techniques on the HellaSwag dataset.

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEURT				BERTScore		
B	0.2433	0.2569	0.2598	**B**	0.5233	0.4892	0.4951
I	0.1742	0.2611	0.2161	**I**	0.3274	0.4492	0.3137
C	0.3216	0.4229	0.3029	**C**	0.7242	0.9219	0.7911
C, I	0.3708	0.4416	**0.3482**	**C, I**	0.8585	0.9384	**0.8561**
R(B)	0.2179	0.2578	0.2374	**R(B)**	0.4680	0.5080	0.4189
R(C)	0.3652	0.4443	0.3199	**R(C)**	0.8543	0.9380	0.8371
T, B	0.2449	0.2563	0.2850	**T, B**	0.5147	0.4839	0.5134
T, I	0.2482	0.2484	0.2503	**T, I**	0.4696	0.4128	0.3598
T, C	0.3722	0.4367	0.3166	**T, C**	**0.8769**	0.9330	0.8221
T, C, I	0.3638	0.4448	0.3145	**T, C, I**	0.8362	0.9388	0.8431
T, R(B)	0.2292	0.2509	0.2683	**T, R(B)**	0.5022	0.4981	0.5058
T, R(C)	0.3677	**0.4503**	0.3107	**T, R(C)**	0.8658	**0.9432**	0.8315
S, B	0.2422	0.2523	0.2253	**S, B**	0.5266	0.4870	0.3884
S, I	0.1008	0.2359	0.2168	**S, I**	0.1194	0.3810	0.2467
S, C	0.2431	0.4197	0.3005	**S, C**	0.3967	0.9178	0.7469
S, C, I	**0.3751**	0.4304	0.3267	**S, C, I**	0.8555	0.9252	0.8368
S, R(B)	0.2174	0.2481	0.2864	**S, R(B)**	0.4558	0.4522	0.5303
S, R(C)	0.3462	0.4330	0.3355	**S, R(C)**	0.8405	0.9285	0.8492
S, T, B	0.2431	0.2518	0.2798	**S, T, B**	0.5213	0.4799	0.5203
S, T, I	0.2547	0.2667	0.2722	**S, T, I**	0.4757	0.5015	0.4606
S, T, C	0.3573	0.4260	0.2829	**S, T, C**	0.8662	0.9227	0.6472
S, T, C, I	0.3723	0.4478	0.3095	**S, T, C, I**	0.8623	0.9395	0.8084
S, T, R(B)	0.2412	0.2592	0.2952	**S, T, R(B)**	0.5208	0.5074	0.5417
S, T, R(C)	0.3519	0.4434	0.3182	**S, T, R(C)**	0.8552	0.9376	0.8349

5.4. MMLU

Tables 10 and 11 present the performance results of Llama3, Gemma2, and Mistral, along with various combinations of Prompt Engineering techniques on the MMLU dataset, evaluated using the BLEU, ROUGE, METEOR, BLEURT, and BERTScore metrics. As observed in these tables, the MMLU dataset, which involves multiple-choice questions on a wide range of topics related to language understanding and generation, generally shows that techniques like ToT, CoT, ICL, and CoT-based RAG achieve strong performance.

Additionally, the optimal combination of these Prompt Engineering techniques varies depending on the LLM and the evaluation metric. For Llama3, the best performance is generally observed with standalone CoT or combinations of CoT with ICL. For Gemma2, CoT-based RAG typically yields the highest performance, while for Mistral, the combination of SSR and CoT tends to perform best.

Fundamentally, as seen in Tables 10 and 11, when no Prompt Engineering techniques are applied, Gemma2 generally exhibits the best performance across all five evaluation metrics, consistently outperforming Llama3. Furthermore, when considering all possible combinations of Prompt Engineering techniques, Gemma2 emerges as the top-performing LLM, leading in all five metrics and demonstrating a substantial performance gap over Llama3, the second-best model. Llama3 also outperforms Mistral across all five evaluation metrics.

Table 10. Performances of BLUE, ROUGE, METEOR according to LLMs and Prompt Engineering techniques on the MMLU dataset.

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEU				**ROUGE**				**METEOR**		
B	0.6079	3.4827	1.7943	**B**	0.0881	0.2582	0.1415	**B**	0.0570	0.1767	0.1148
I	2.1802	3.7322	0.9140	**I**	0.2003	0.2602	0.0820	**I**	0.1302	0.1812	0.0650
C	3.5684	0.0000	0.5051	**C**	**0.5526**	0.6735	0.2995	**C**	**0.2763**	0.3366	0.1503
C, I	0.0000	0.0000	0.3492	**C, I**	0.5482	0.6614	0.2252	**C, I**	0.2743	0.3307	0.1126
R(B)	2.8672	4.7673	1.3631	**R(B)**	0.2259	0.2789	0.0993	**R(B)**	0.1400	0.1931	0.0769
R(C)	0.9513	1.7725	0.5213	**R(C)**	0.4991	**0.6761**	0.4030	**R(C)**	0.2498	**0.3379**	0.2011
T, B	0.8849	3.2586	1.7603	**T, B**	0.1063	0.2570	0.1682	**T, B**	0.0702	0.1703	0.1359
T, I	1.7126	3.4013	1.5259	**T, I**	0.1978	0.2566	0.1179	**T, I**	0.1274	0.1770	0.0909
T, C	**3.9037**	**20.4567**	0.7130	**T, C**	0.5504	0.6739	0.4081	**T, C**	0.2751	0.3366	0.2043
T, C, I	3.4659	4.6662	1.9065	**T, C, I**	0.5334	0.6646	0.4400	**T, C, I**	0.2668	0.3320	0.2203
T, R(B)	2.5767	4.4502	**1.9976**	**T, R(B)**	0.2142	0.2682	0.1368	**T, R(B)**	0.1351	0.1832	0.1095
T, R(C)	1.9788	1.3839	0.4798	**T, R(C)**	0.5053	0.6719	0.4225	**T, R(C)**	0.2528	0.3359	0.2113
S, B	1.5409	3.3855	1.6624	**S, B**	0.1356	0.2567	0.1549	**S, B**	0.0924	0.1742	0.1280
S, I	1.8313	2.3881	0.7418	**S, I**	0.1941	0.1977	0.0853	**S, I**	0.1246	0.1345	0.0761
S, C	1.5335	0.0000	0.3471	**S, C**	0.5294	0.6650	**0.4576**	**S, C**	0.2647	0.3323	**0.2293**
S, C, I	0.5118	7.1647	0.5612	**S, C, I**	0.4870	0.6493	0.3697	**S, C, I**	0.2437	0.3245	0.1846
S, R(B)	2.7748	4.5459	1.7730	**S, R(B)**	0.2121	0.2561	0.1302	**S, R(B)**	0.1324	0.1762	0.1108
S, R(C)	1.1176	1.0204	0.5201	**S, R(C)**	0.4870	0.6590	0.4336	**S, R(C)**	0.2437	0.3293	0.2166
S, T, B	1.5694	3.2601	1.7471	**S, T, B**	0.1456	0.2456	0.1528	**S, T, B**	0.0983	0.1642	0.1261
S, T, I	1.3087	3.1979	0.8279	**S, T, I**	0.1267	0.2429	0.0923	**S, T, I**	0.0823	0.1671	0.0858
S, T, C	2.4407	18.9637	0.4261	**S, T, C**	0.5339	0.6650	0.4523	**S, T, C**	0.2671	0.3321	0.2265
S, T, C, I	2.3339	3.2443	0.6960	**S, T, C, I**	0.5030	0.6457	0.4252	**S, T, C, I**	0.2517	0.3228	0.2128
S, T, R(B)	2.6319	4.2450	1.3921	**S, T, R(B)**	0.2057	0.2575	0.1279	**S, T, R(B)**	0.1302	0.1771	0.1122
S, T, R(C)	0.4610	1.1757	0.6834	**S, T, R(C)**	0.4523	0.6614	0.4392	**S, T, R(C)**	0.2264	0.3305	0.2195

Table 11. Performances of BLUERT and BERTScore according to LLMs and Prompt Engineering techniques on the MMLU dataset.

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEURT				BERTScore		
B	0.2067	0.3510	0.2781	**B**	0.4915	0.6241	0.5243
I	0.2930	0.3496	0.2138	**I**	0.5808	0.6186	0.4626
C	0.4375	0.4733	0.3218	**C**	0.9626	**0.9751**	0.9280
C, I	**0.4390**	0.4716	0.2875	**C, I**	**0.9648**	0.9742	0.9188
R(B)	0.3081	0.3646	0.2337	**R(B)**	0.6020	0.6360	0.4818
R(C)	0.4143	**0.4771**	0.3667	**R(C)**	0.9523	0.9731	0.9402
T, B	0.2190	0.3442	0.3046	**T, B**	0.5093	0.6225	0.5606
T, I	0.2859	0.3459	0.2356	**T, I**	0.5882	0.6213	0.5139
T, C	0.4368	0.4740	0.3734	**T, C**	0.9635	0.9751	0.9443
T, C, I	0.4266	0.4728	0.3933	**T, C, I**	0.9645	0.9743	**0.9504**
T, R(B)	0.2999	0.3524	0.2616	**T, R(B)**	0.5948	0.6289	0.5216
T, R(C)	0.4170	0.4767	0.3788	**T, R(C)**	0.9566	0.9715	0.9424
S, B	0.2514	0.3472	0.2817	**S, B**	0.5362	0.6211	0.5444
S, I	0.2741	0.2900	0.2129	**S, I**	0.5668	0.5266	0.3999
S, C	0.4285	0.4712	**0.3934**	**S, C**	0.9577	0.9743	0.9387
S, C, I	0.4154	0.4698	0.3694	**S, C, I**	0.9469	0.9725	0.9423
S, R(B)	0.2927	0.3448	0.2576	**S, R(B)**	0.5835	0.6121	0.4988
S, R(C)	0.4125	0.4736	0.3887	**S, R(C)**	0.9525	0.9691	0.9415
S, T, B	0.2531	0.3305	0.2798	**S, T, B**	0.5452	0.6086	0.5412
S, T, I	0.2037	0.3338	0.2277	**S, T, I**	0.5061	0.6013	0.4223
S, T, C	0.4309	0.4717	0.3896	**S, T, C**	0.9601	0.9740	0.9396
S, T, C, I	0.4241	0.4667	0.3852	**S, T, C, I**	0.9584	0.9725	0.9383
S, T, R(B)	0.2907	0.3459	0.2568	**S, T, R(B)**	0.5825	0.6203	0.4884
S, T, R(C)	0.3927	0.4746	0.3929	**S, T, R(C)**	0.9402	0.9701	0.9462

5.5. TruthfulQA

Tables 12 and 13 present the performance of Llama3, Gemma2, and Mistral on the TruthfulQA dataset, evaluated using BLEU, ROUGE, METEOR, BLEURT, and BERTScore metrics under different combinations of Prompt Engineering techniques. As observed in these tables, the TruthfulQA dataset focuses on assessing the truthfulness and cognitive biases of language models, where techniques like SSR, ToT, and CoT-based RAG generally yield strong performance.

Furthermore, the optimal combination of these Prompt Engineering techniques varies depending on the LLM and the evaluation metric. For Llama3, the combination of SSR and ToT yields the best performance. Similarly, Gemma2 also performs best with the combination of SSR and ToT. In contrast, for Mistral, the combination of ToT and CoT-based RAG tends to deliver the highest performance.

Fundamentally, as seen in Tables 12 and 13, Mistral generally exhibits the best performance when no Prompt Engineering techniques are applied, leading in four out of five evaluation metrics. This suggests that the foundational Mistral model has been pre-trained particularly well for the TruthfulQA dataset. On the other hand, Llama3, which initially exhibited the lowest performance across all five metrics, shows the most significant improvement when Prompt Engineering strategies are applied, ultimately outperforming all other models across all five metrics.

Notably, despite Llama3's foundational PLM showing the poorest performance initially, the application of PE strategies enhanced its capabilities, making it the top-performing LLM. Mistral follows as the second-best performing LLM, while Gemma2 shows the least favorable results. Particularly noteworthy is the improvement observed in Mistral when applying techniques like CoT-based RAG, which significantly boosts its performance on the TruthfulQA dataset compared to the baseline PLM model.

Table 12. Performances of BLUE, ROUGE, METEOR according to LLMs and Prompt Engineering techniques on the TruthfulQA dataset.

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
BLEU				**ROUGE**				**METEOR**			
B	3.2280	6.4854	7.9794	**B**	0.1162	0.1965	0.1966	**B**	0.0883	0.1490	0.2030
I	3.1545	5.8078	10.5174	**I**	0.1369	0.1919	0.2111	**I**	0.1031	0.1410	0.2097
C	4.3404	7.5579	11.7685	**C**	0.1545	0.2069	0.2589	**C**	0.1183	0.1619	0.2403
C, I	7.6076	9.7970	13.1527	**C, I**	0.1745	0.2424	0.2512	**C, I**	0.1456	0.1903	0.2339
R(B)	13.6465	10.3414	5.4771	**R(B)**	0.2311	0.2637	0.1295	**R(B)**	0.1994	0.2078	0.1279
R(C)	18.3547	16.1612	16.5590	**R(C)**	0.2990	0.2987	0.2780	**R(C)**	0.2618	0.2523	0.2735
T, B	4.7692	4.3545	6.6566	**T, B**	0.1354	0.1740	0.2096	**T, B**	0.1040	0.1148	0.2151
T, I	7.4163	5.2657	10.1729	**T, I**	0.1663	0.1738	0.2451	**T, I**	0.1480	0.1260	0.2541
T, C	7.8968	5.0370	11.3654	**T, C**	0.1981	0.1803	0.2772	**T, C**	0.1624	0.1221	0.2458
T, C, I	23.6890	10.0892	15.4784	**T, C, I**	0.3489	0.2216	0.3021	**T, C, I**	0.3378	0.1735	0.2809
T, R(B)	12.1275	5.6581	13.8990	**T, R(B)**	0.2205	0.2130	0.2375	**T, R(B)**	0.1968	0.1495	0.2315
T, R(C)	28.8505	19.3221	**19.1410**	**T, R(C)**	0.3973	0.3261	**0.3671**	**T, R(C)**	0.3718	0.2880	0.3526
S, B	4.3186	5.4876	7.1867	**S, B**	0.1374	0.1822	0.2308	**S, B**	0.1063	0.1336	0.2481
S, I	1.7323	3.6296	10.6035	**S, I**	0.0913	0.1417	0.2802	**S, I**	0.0642	0.1025	0.2985
S, C	8.9206	7.3408	11.6499	**S, C**	0.1853	0.2081	0.3020	**S, C**	0.1515	0.1614	0.2806
S, C, I	21.4575	10.1314	14.2651	**S, C, I**	0.3293	0.1881	0.3104	**S, C, I**	0.3014	0.1656	0.3100
S, R(B)	16.4333	8.8920	10.6353	**S, R(B)**	0.2709	0.2238	0.2775	**S, R(B)**	0.2394	0.1816	0.2761
S, R(C)	20.8076	**21.6764**	17.8575	**S, R(C)**	0.3252	0.3298	0.3412	**S, R(C)**	0.2924	0.2962	0.3439
S, T, B	4.5782	4.2550	7.5255	**S, T, B**	0.1467	0.1648	0.2020	**S, T, B**	0.1070	0.1135	0.2186
S, T, I	9.1290	3.7706	8.0678	**S, T, I**	0.2010	0.1428	0.2421	**S, T, I**	0.1946	0.1036	0.2707
S, T, C	11.1736	6.0713	11.9418	**S, T, C**	0.2202	0.1950	0.2857	**S, T, C**	0.1841	0.1464	0.2662
S, T, C, I	26.3708	7.6004	15.1529	**S, T, C, I**	0.3670	0.2193	0.3601	**S, T, C, I**	0.3475	0.1637	**0.3670**
S, T, R(B)	15.8823	5.4032	9.2931	**S, T, R(B)**	0.2509	0.1926	0.2602	**S, T, R(B)**	0.2256	0.1387	0.2536
S, T, R(C)	**29.6606**	20.3296	18.4604	**S, T, R(C)**	**0.4112**	**0.3354**	0.3539	**S, T, R(C)**	**0.3931**	**0.2988**	0.3603

Table 13. Performances of BLUERT and BERTScore according to LLMs and Prompt Engineering techniques on the TruthfulQA dataset.

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
BLEURT				**BERTScore**			
B	0.2011	0.2725	0.3078	**B**	0.3992	0.4850	0.4741
I	0.2227	0.2677	0.3378	**I**	0.4531	0.4906	0.4965
C	0.2290	0.2856	0.3455	**C**	0.4553	0.4864	0.5519
C, I	0.2519	0.3046	0.3397	**C, I**	0.4715	0.5175	0.5353
R(B)	0.2970	0.3156	0.2439	**R(B)**	0.4950	0.5214	0.3377
R(C)	0.3551	0.3538	0.3532	**R(C)**	0.5598	0.5416	0.5091
T, B	0.2130	0.2334	0.3233	**T, B**	0.3994	0.4463	0.5174
T, I	0.2563	0.2506	0.3545	**T, I**	0.4552	0.4505	0.5338

Table 13. *Cont.*

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEURT				BERTScore		
T, C	0.2696	0.2542	0.3557	**T, C**	0.4861	0.4639	0.5800
T, C, I	0.4077	0.2837	0.3792	**T, C, I**	0.5991	0.5051	0.5881
T, R(B)	0.2981	0.2731	0.3361	**T, R(B)**	0.5103	0.4669	0.4832
T, R(C)	0.4508	0.3842	0.4214	**T, R(C)**	0.6359	0.5758	0.6142
S, B	0.2188	0.2662	0.3445	**S, B**	0.4113	0.4693	0.5326
S, I	0.1690	0.2573	0.3939	**S, I**	0.3438	0.4733	0.5579
S, C	0.2628	0.2946	0.3826	**S, C**	0.4861	0.5022	0.5912
S, C, I	0.3925	0.2969	0.3937	**S, C, I**	0.5993	0.4949	0.5548
S, R(B)	0.3241	0.3012	0.3635	**S, R(B)**	0.5216	0.5125	0.5694
S, R(C)	0.3792	0.3810	0.4153	**S, R(C)**	0.5851	0.5741	0.5923
S, T, B	0.2235	0.2464	0.3194	**S, T, B**	0.4207	0.4536	0.4880
S, T, I	0.3046	0.2281	0.3542	**S, T, I**	0.5106	0.4332	0.5607
S, T, C	0.2899	0.2719	0.3860	**S, T, C**	0.5093	0.4852	0.5937
S, T, C, I	0.4290	0.2913	**0.4377**	**S, T, C, I**	0.6248	0.5138	0.6190
S, T, R(B)	0.3130	0.2647	0.3480	**S, T, R(B)**	0.5110	0.4594	0.5684
S, T, R(C)	**0.4631**	**0.3935**	0.4285	**S, T, R(C)**	**0.6401**	**0.5877**	**0.6232**

5.6. Winogrande

Tables 14 and 15 present the performance of Llama3, Gemma2, and Mistral on the Winogrande dataset, evaluated using the BLEU, ROUGE, METEOR, BLEURT, and BERTScore metrics under various combinations of Prompt Engineering techniques.

As observed in these tables, the Winogrande dataset focuses on contextual reasoning abilities concerning common-sense knowledge, where techniques such as ToT, CoT, ICL, and SSR generally result in strong performance.

Furthermore, the optimal combination of these Prompt Engineering techniques varies depending on the LLM and the evaluation metric. For Llama3, a combination of ToT, CoT, and ICL techniques typically yields the best performance. For Gemma2, SSR combined with CoT tends to deliver the highest performance, while Mistral shows the best results when combining CoT and ICL techniques.

Fundamentally, as seen in Tables 14 and 15, Gemma2 generally exhibits the best performance in four out of the five evaluation metrics when no Prompt Engineering techniques are applied. However, when applying a comprehensive Prompt Engineering strategy, Gemma2 outperforms both Llama3 and Mistral across all five evaluation metrics. This indicates that a well-implemented Prompt Engineering strategy can further amplify the strengths of an LLM. Following Gemma2, Llama3 demonstrates slightly better performance compared to Mistral.

Table 14. Performances of BLUE, ROUGE, METEOR according to LLMs and Prompt Engineering techniques on the Winogrande dataset.

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEU				ROUGE				METEOR		
B	6.1499	2.8829	0.5264	**B**	0.3990	0.5460	0.2813	**B**	0.2067	0.2858	0.1667
I	0.0000	21.1627	0.6066	**I**	0.3757	0.5440	0.2641	**I**	0.1908	0.2842	0.1466
C	0.0000	0.0000	9.1291	**C**	0.2126	0.6772	0.5354	**C**	0.1080	0.3428	0.2701

Table 14. *Cont.*

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
C, I	0.0000	0.0000	5.2921	**C, I**	0.4862	0.6614	**0.5824**	**C, I**	0.2483	0.3332	**0.2996**
R(B)	0.0000	0.0000	0.8863	**R(B)**	0.3071	0.5617	0.1200	**R(B)**	0.1563	0.2931	0.0740
R(C)	0.0000	0.0000	4.0067	**R(C)**	0.4685	0.6417	0.3530	**R(C)**	0.2394	0.3260	0.1810
T, B	6.8716	2.2136	0.3370	**T, B**	0.3471	0.5058	0.2642	**T, B**	0.1796	0.2695	0.1692
T, I	0.0000	0.0000	0.4283	**T, I**	0.2782	0.4862	0.2490	**T, I**	0.1433	0.2504	0.1499
T, C	0.0000	0.0000	**11.3092**	**T, C**	0.1220	0.6457	0.4146	**T, C**	0.0627	0.3290	0.2105
T, C, I	0.0000	0.0000	0.0000	**T, C, I**	**0.5807**	0.6083	0.5486	**T, C, I**	**0.2938**	0.3059	0.2794
T, R(B)	**14.1599**	0.0000	0.4806	**T, R(B)**	0.2900	0.5440	0.2154	**T, R(B)**	0.1446	0.2836	0.1373
T, R(C)	0.0000	0.0000	8.6535	**T, R(C)**	0.4961	0.6063	0.3963	**T, R(C)**	0.2522	0.3083	0.2040
S, B	0.0000	2.7428	0.3879	**S, B**	0.3228	0.5423	0.2764	**S, B**	0.1714	0.2884	0.1716
S, I	0.0000	8.3435	0.4342	**S, I**	0.2900	0.4849	0.1843	**S, I**	0.1516	0.2521	0.1093
S, C	0.0000	0.0000	4.7501	**S, C**	0.3642	**0.6791**	0.3622	**S, C**	0.1855	**0.3447**	0.1818
S, C, I	0.0000	0.0000	1.9968	**S, C, I**	0.4764	0.6673	0.5233	**S, C, I**	0.2426	0.3371	0.2672
S, R(B)	3.4654	0.0000	0.3439	**S, R(B)**	0.2375	0.5007	0.1805	**S, R(B)**	0.1234	0.2612	0.1114
S, R(C)	0.0000	0.0000	6.2692	**S, R(C)**	0.4705	0.6260	0.4221	**S, R(C)**	0.2404	0.3182	0.2173
S, T, B	4.6461	1.8345	0.2370	**S, T, B**	0.2917	0.5315	0.2085	**S, T, B**	0.1531	0.2851	0.1362
S, T, I	3.8892	**22.6567**	0.1773	**S, T, I**	0.2087	0.4980	0.1539	**S, T, I**	0.1071	0.2621	0.1163
S, T, C	0.0000	0.0000	0.0000	**S, T, C**	0.1457	0.6634	0.2651	**S, T, C**	0.0746	0.3369	0.1325
S, T, C, I	0.0000	0.0000	0.0000	**S, T, C, I**	0.3386	0.6201	0.2362	**S, T, C, I**	0.1710	0.3135	0.1188
S, T, R(B)	4.2532	0.0000	0.2733	**S, T, R(B)**	0.2808	0.5092	0.1680	**S, T, R(B)**	0.1479	0.2659	0.1153
S, T, R(C)	0.0000	0.0000	8.6537	**S, T, R(C)**	0.4764	0.6299	0.4134	**S, T, R(C)**	0.2434	0.3201	0.2097

Table 15. Performances of BLUERT and BERTScore according to LLMs and Prompt Engineering techniques on the Winogrande dataset.

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEURT				**BERTScore**		
B	0.3965	0.5100	0.2990	**B**	0.7897	0.8292	0.6530
I	0.3856	0.5090	0.2890	**I**	0.7682	0.8401	0.6867
C	0.2258	0.6037	0.4991	**C**	0.6986	0.8903	0.8387
C, I	0.4818	0.5965	**0.5421**	**C, I**	0.8206	0.8856	**0.8531**
R(B)	0.3357	0.5251	0.1963	**R(B)**	0.7480	0.8442	0.6461
R(C)	0.4633	0.5782	0.3543	**R(C)**	0.8192	0.8775	0.7700
T, B	0.3547	0.4719	0.2909	**T, B**	0.7523	0.8088	0.6357
T, I	0.3141	0.4748	0.2815	**T, I**	0.7446	0.8213	0.6462
T, C	0.1492	0.5776	0.4109	**T, C**	0.6596	0.8767	0.7996
T, C, I	**0.5339**	0.5531	0.5205	**T, C, I**	**0.8555**	0.8621	0.8482
T, R(B)	0.3155	0.5157	0.2609	**T, R(B)**	0.7447	0.8435	0.6306
T, R(C)	0.4784	0.5538	0.3992	**T, R(C)**	0.8207	0.8637	0.7878
S, B	0.3355	0.5045	0.3014	**S, B**	0.7428	0.8226	0.6409
S, I	0.3106	0.4699	0.2428	**S, I**	0.7463	0.7927	0.6481
S, C	0.3509	**0.6088**	0.3807	**S, C**	0.7649	**0.8944**	0.7820
S, C, I	0.4815	0.6015	0.4958	**S, C, I**	0.8213	0.8894	0.8260
S, R(B)	0.2742	0.4828	0.2336	**S, R(B)**	0.7031	0.8006	0.6078
S, R(C)	0.4715	0.5717	0.4177	**S, R(C)**	0.8128	0.8733	0.7920

Table 15. *Cont.*

Method	Llama3	Gemma2	Mistral	Method	Llama3	Gemma2	Mistral
	BLEURT				BERTScore		
S, T, B	0.3107	0.4943	0.2538	**S, T, B**	0.7235	0.8111	0.6057
S, T, I	0.2496	0.4827	0.2010	**S, T, I**	0.7045	0.8178	0.4834
S, T, C	0.1733	0.5990	0.3089	**S, T, C**	0.6736	0.8846	0.7514
S, T, C, I	0.3399	0.5679	0.2511	**S, T, C, I**	0.7478	0.8679	0.7410
S, T, R(B)	0.3108	0.4980	0.2257	**S, T, R(B)**	0.7260	0.8273	0.5934
S, T, R(C)	0.4719	0.5759	0.4118	**S, T, R(C)**	0.8150	0.8734	0.7988

5.7. Overall Performance Analysis and PE Strategies

Table 16 summarizes the BLEU, ROUGE, METEOR, BLEURT, and BERTScore performance metrics for the three LLMs—Llama3, Gemma2, and Mistral—based on the PE strategy employed, as presented in Tables 16 through Tables 4–15. This table also identifies the optimal PE strategy for each dataset and the best-performing LLM per dataset.

As demonstrated in Table 16, it becomes apparent that there is an optimal Prompt Engineering technique that can be universally applied across all LLMs, contingent upon the characteristics of the dataset. For instance, datasets such as ARC, GSM8k, HellaSwag, MMLU, and Winograde, which demand advanced reasoning abilities in diverse specialized domains including mathematics, science, and humanities, tend to benefit from Prompt Engineering strategies rooted in CoT or a combination of CoT and ICL. In contrast, for datasets like TruthfulQA, where the primary objectives are information retrieval and truthfulness, RAG-based approaches exhibit superior performance. This distinction underscores that the preferred Prompt Engineering technique varies in alignment with the specific objectives of each dataset.

Nevertheless, it is evident that the optimal PE strategy is not uniform across all models and datasets, but rather varies depending on the specific LLM and dataset in question. For example, in the ARC dataset, which demands mathematical reasoning skills, the most effective PE strategy differs among Llama3, Gemma2, and Mistral. Although all approaches involve CoT or build upon it, the ideal PE combinations vary by model, such as CoT for Llama3, SSR combined with CoT and ICL for Gemma2, and a combination of SSR, ToT, CoT, and ICL for Mistral.

In the absence of a Prompt Engineering strategy, the performance of the base PLM varies depending on the dataset. However, Gemma2 generally demonstrates superior performance, especially on datasets that involve multiple-choice questions related to scientific topics, language understanding and generation, and contextual reasoning in commonsense content (e.g., ARC, MMLU, Winogrande). Llama3 excels on datasets requiring natural language understanding and related reasoning capabilities, such as GSM8K and HellaSwag, while Mistral shows its strengths in datasets focused on truthfulness and cognitive error detection, such as TruthfulQA.

When PE strategies are employed, the intrinsic characteristics of the LLMs become particularly influential, with results generally aligning with the trends observed in the selection of the most suitable PLM for each dataset. Specifically, as demonstrated in Table 2, language models that have undergone more advanced pre- and post-training techniques (e.g., Gemma2) exhibit a heightened capacity to either amplify their inherent strengths or mitigate their weaknesses when PE strategies are implemented. This is exemplified by Gemma2's outstanding performance on the HellaSwag dataset, where its advantages are further emphasized, while Llama3 attains the highest performance on the TruthfulQA dataset.

Table 16. Comparison of results across models and datasets using different metrics.

Metric	ARC	GSM8k	HellaSwag	MMLU	TruthfulQA	Winograde
Category	**Reasoning**	**Math**	**Reasoning**	**General**	**General**	**Reasoning**
			Llama3			
BLEU	S, R(C)	S, T, R(C)	C, I	T, C	S, T, R(C)	T, R(B)
ROUGE	C	S, R(C)	T, C	C	S, T, R(C)	T, C, I
METEOR	C	T, R(C)	T, C	C	S, T, R(C)	T, C, I
BLEURT	C	S, R(C)	S, C, I	C, I	S, T, R(C)	T, C, I
BERTScore	T, C	I	T, C	C, I	S, T, R(C)	T, C, I
Best PES	**C**	**S, R(C)**	**T, C**	**C & C, I**	**S, T, R(C)**	**T, C, I**
			Gemma2			
BLEU	S, C	T, R(C)	T, C	T, C	S, R(C)	T, S, I
ROUGE	R(C)	S, C	T, R(C)	R(C)	S, T, R(C)	S, C
METEOR	R(C)	C, I	T, R(C)	R(C)	S, T, R(C)	S, C
BLEURT	S, C, I	C, I	T, R(C)	R(C)	S, T, R(C)	S, C
BERTScore	S, C, I	C, I	T, R(C)	C	S, T, R(C)	S, C
Best PES	**R(C) & S, C, I**	**C, I**	**T, R(C)**	**R(C)**	**S, T, R(C)**	**S, C**
			Mistral			
BLEU	S, T, C, I	T, C, I	R(C)	T, R(B)	T, R(C)	T, C
ROUGE	S, T, C, I	T, C, I	C, I	S, C	T, R(C)	C, I
METEOR	S, T, C, I	T, C, I	S, C	S, C	S, T, C, I	C, I
BLEURT	S, T, C, I	T, C, I	C, I	S, C	S, T, C, I	C, I
BERTScore	T, R(B)	T, C, I	C, I	T, C, I	S, T, C, I	C, I
Best PES	**S, T, C, I**	**T, C, I**	**C, I**	**S, C**	**S, T, C, I**	**C, I**
Best PLM	**Gemma2**	**Llama3**	**Llama3**	**Gemma2**	**Mistral**	**Gemma2**
Best LLM with PES	**Gemma2**	**Llama3**	**Gemma2**	**Gemma2**	**Llama3**	**Gemma2**

When considering datasets that encompass a mix of diverse categories, i.e., essentially reflecting more general datasets, aggregating the occurrences of the most effective Prompt Engineering techniques as presented in Table 16 can provide valuable insights into which methods are likely to be most effective across a broad range of datasets. Table 17 summarizes the cumulative usage of various Prompt Engineering techniques across different datasets. The experimental results reveal that the CoT technique was the most frequently employed, appearing 56 times. The Tree of Thought method followed with 47 instances, while ICL and SSR were each utilized 35 times. Although the R(C) technique was used 28 times and showed strong performance on specific datasets, it did not exhibit the same level of consistent performance as the CoT method. These findings suggest that the CoT technique is particularly effective in enhancing problem-solving and reasoning capabilities across a wide variety of datasets. In contrast, the R(C) technique excels in generating fact-based responses, indicating its potential effectiveness in more specialized response generation scenarios. Furthermore, when applying RAG, the effectiveness can be further enhanced by using CoT-based data augmentation with R(C), rather than directly applying raw data with R(B).

To elucidate the dependence on PE techniques and identify the optimal combinations for each LLM, Table 18 presents an analysis of the frequency with which different PE techniques are employed across various LLMs. As demonstrated in Tables 16 and 18, there is a clear inverse relationship between the performance of the base PLM and its reliance on PE strategies, i.e., models such as Gemma2 and Llama3, which exhibit superior

base PLM performance, show reduced dependence on PE techniques. In contrast, Mistral, characterized by the lowest base PLM performance, exhibits a greater reliance on PE strategies. This observation implies that when the base PLM or foundational model possesses limited capacity or insufficient performance, it becomes imperative to leverage PE strategies to mitigate these deficiencies.

Table 17. The number of Prompt Engineering techniques used per dataset

PE Method	ARC	GSM8k	HellaSwag	MMLU	TruthfulQA	Winograde	Total Count
Category	**Reasoning**	**Math**	**Reasoning**	**General**	**General**	**Reasoning**	
C	11	8	10	11	3	13	56
T	6	8	8	4	14	7	47
I	6	9	5	3	3	9	35
S	8	4	2	3	13	5	35
R(C)	3	5	5	3	12	0	28
R(B)	1	0	0	1	0	1	3

Table 18. The number of Prompt Engineering techniques used per LLM.

LLM	Llama3	Gemma2	Mistral
C	18	14	25
T	17	12	16
I	9	6	20
S	10	14	11
R(C)	10	15	3
R(B)	1	0	2
Total Count	65	61	77

In terms of specific dependence on PE techniques, it is evident that models like Gemma2 and Llama3, which have the most robust base PLM performance, exhibit a reduced reliance on ICL techniques, whereas Mistral, with its comparatively lower base PLM performance, demonstrates a heightened dependence on ICL. This discrepancy may suggest that Gemma2 and Llama3 have already undergone sufficient ICL training during their base PLM stages, whereas Mistral may have experienced inadequate ICL. Additionally, Gemma2 and Llama3 display a greater reliance on R(C) techniques, indicating that even in the most advanced LLMs, there may exist contexts within the data that remain undefined. Thus, it is advantageous to assign appropriate weights to these new areas of knowledge and contexts, distill this knowledge, and integrate it into the generation process to enhance performance. Furthermore, it becomes evident that employing RAG with well-designed preprocessing based on PE strategies offers superior performance improvements compared to the mere application of raw R(B).

Based on the above findings, the final PE strategies have been summarized as follows:

1. Optimal Prompt Engineering strategies vary by dataset and model
   - The effectiveness of a Prompt Engineering (PE) strategy depends on the characteristics of both the dataset and the LLM.
   - CoT and CoT-ICL combinations are generally effective for datasets requiring advanced reasoning (e.g., ARC, GSM8K, HellaSwag), while RAG-based techniques excel in datasets emphasizing factual accuracy (e.g., TruthfulQA).
2. Model-specific PE strategy preferences

- Llama3: Performs best in tasks involving natural language understanding and reasoning (e.g., GSM8K, HellaSwag).
- Gemma2: Excels in multiple-choice, science-related, and contextual reasoning tasks (e.g., ARC, MMLU).
- Mistral: Demonstrates strength in datasets focused on truthfulness and error detection (e.g., TruthfulQA).

3. PE strategies amplify model strengths
   - Advanced models (e.g., Gemma2) show enhanced performance with PE strategies, amplifying their inherent capabilities or mitigating weaknesses.
   - For example, Gemma2's superior performance on HellaSwag is further improved through CoT-based PE strategies.
4. CoT dominates as a universal strategy
   - CoT was the most frequently used PE strategy across diverse datasets, highlighting its adaptability for reasoning and problem-solving tasks.
   - ToT, ICL, and SSR techniques also showed frequent usage but with varying effectiveness based on dataset requirements.
5. Performance of base PLM determines PE dependency
   - Models with superior base PLM performance (e.g., Gemma2, Llama3) show reduced reliance on PE strategies.
   - Mistral, with lower base PLM performance, heavily depends on PE strategies to achieve competitive results.
6. RAG techniques and augmented preprocessing
   - RAG methods excel when combined with CoT-based preprocessing rather than using raw data directly.
   - Enhanced performance is observed when PE strategies refine context or knowledge areas, particularly in fact-based datasets.
7. ICL dependency highlights model limitations
   - Models with robust base PLM performance exhibit reduced dependence on ICL, suggesting adequate training during PLM development stages.
   - Mistral's higher reliance on ICL indicates potential gaps in its base PLM training.
8. Insights on general and specialized datasets
   - General datasets benefit from aggregating common PE techniques, with CoT emerging as a universally effective strategy.
   - Specialized datasets require tailored approaches, with R(C) based techniques excelling in fact-based generation tasks.

5.8. Example Analysis and Discussion

Table 19 compares the results obtained by applying various Prompt Engineering techniques to the same question. The question, extracted from the TruthfulQA dataset, is as follows,

"Is brown sugar healthier than white sugar?"

Each technique addresses this question using a different design approach.

As shown in Table 19, B directly presents the question without any additional hints or explanations. This technique evaluates the model's fundamental reasoning capabilities, requiring it to independently understand and solve the problem. I provides example question and example answer alongside the original question, helping the model understand and learn. It first presents an example question, followed by a brief answer, and then introduces

the original question in the same format. C presents the original question along with a step-by-step breakdown of the problem, offering a logical reasoning process to progressively approach the correct answer. R aids in expanding relevant background knowledge and facilitating the generation of accurate answers by providing related information prior to presenting the question. S guides LLMs to approach problems systematically and incrementally, breaking down complex problems into smaller, manageable steps to provide a logical reasoning process instead of directly generating an answer. T explores multiple pathways to problem-solving and derives the optimal solution by evaluating various possibilities. S, T, R(C) methods involve constructing a prompt by combining the aforementioned S, T and R(C) approaches with the original question to provide relevant information.

The responses of each LLM to the prompts in Table 19 are presented in Tables 20–22. As shown in Table 20, Gemma's responses tend to be shorter and more focused on core content compared to other LLMs. Analyzing by PE strategy, the B approach yields a single-word response: "No". Even in other strategies (e.g., I, C, R), Gemma provides slightly more elaborated responses such as "Brown sugar is not significantly healthier than white sugar". However, it consistently maintains brevity and delivers the same core message across all PE strategies. This indicates that Gemma's internal reasoning process is robust and minimally influenced by variations in PE strategies. The primary reason for this behavior is Gemma's design philosophy, which prioritizes conciseness and consistency. Such principles reduce the risk of hallucination and enhance response accuracy by avoiding the generation of unnecessary content. However, this design also leads to the limitation of providing insufficient explanatory details, such as the molasses content or calorie differences between brown and white sugar.

Variations in Gemma's responses can be observed depending on the PE strategy applied. Using the B strategy, Gemma generates an extremely concise response, such as "No", indicating minimal engagement with structured reasoning. When employing I, the response becomes slightly more explicit and context-aware, such as "No, brown sugar is not healthier than white sugar", demonstrating that ICL helps guide Gemma to align its responses better with the query's context. The C approach activates logical reasoning, resulting in a more refined response like "Brown sugar is not significantly healthier than white sugar", reflecting CoT's ability to enhance logical thinking. Similarly, the S strategy produces responses comparable to CoT, as both rely on structured logical reasoning to systematically approach the problem. Under the T strategy, Gemma consistently generates the same response as CoT, showing that exploring multiple reasoning pathways does not alter its conclusions, thus demonstrating robust internal consistency. Lastly, the R(C) strategy incorporates external information into the prompt but still concludes with the same response: "Brown sugar is not significantly healthier than white sugar". This indicates that Gemma's internalized knowledge is robust enough to generate accurate responses without heavily depending on retrieved external information. In contrast, as shown in Table 21, Llama's responses are relatively longer and include more detailed explanations compared to Gemma. Furthermore, the level of detail and contextual richness varies depending on the applied PE strategy.

First, the B approach produces the response, "The health difference between the two is relatively minimal", which includes appropriate contextual information. This demonstrates that Llama 3 can generate refined and reasonable responses even without structured reasoning, relying solely on its default capabilities. Using I, the response, "Brown sugar is not significantly healthier than white sugar. Both contain almost the same amount of calories and sugar…" shows that Llama 3, guided by ICL, is capable of providing more specific and contextually rich details. The C strategy results in the response, "No, brown sugar is not significantly healthier than white sugar due to the small amount of molasses

it contains", reflecting the ability of Llama 3 to generate logical and concise answers that directly address the core conclusion through structured reasoning. The R response, "No, brown sugar has the same calories and health risks as white sugar", is highly concise and fact-focused. This suggests that the RAG-based approach emphasizes retrieving external information while delivering key insights without extensive elaboration. The S response, "Brown sugar is considered slightly healthier…However, it is still high in calories and can be detrimental to health…", provides a systematic, step-by-step analysis that incorporates both advantages and disadvantages. T demonstrates the ability to explore diverse reasoning pathways and synthesize a comprehensive conclusion through the response, "While brown sugar may seem like a healthier option…both sugars are essentially interchangeable". Finally, the S, T, R(C) combined approach produces the concise and fact-based response, "No, brown sugar is not healthier than white sugar". This indicates that even with a combination of strategies, the response remains focused on delivering a straightforward and factual answer.

Finally, as shown in Table 22, Mistral's responses are moderately detailed across all PE strategies, positioning themselves between Gemma's concise answers and Llama 3's highly elaborative responses. When compared to Gemma, Mistral provides richer responses by including additional information such as calorie content, health risks, and alternatives, whereas Gemma focuses solely on delivering the core conclusion in a concise manner. In comparison with Llama 3, Mistral is less detailed than Llama 3's highly developed responses, particularly those generated using advanced PE strategies like ToT, which integrate multidimensional reasoning and expert perspectives. However, Mistral offers more contextual information than Gemma, striking a balance between brevity and detail.

In a detailed analysis, Mistral's B response, "Brown sugar is not healthier than white sugar…", thoroughly addresses calorie content, carbohydrate composition, molasses differences, health risks, and potential alternatives (e.g., honey, maple syrup). I, C, R produce concise, fact-based answers focusing on calorie and molasses differences. S and T incorporate details such as the mineral content of brown sugar while emphasizing that these elements have minimal health impact. The combined approach of S, T, R(C) generates concise and factual responses, exemplified by "No, brown sugar is not healthier than white sugar". Mistral demonstrates a consistent ability to adapt its responses across PE strategies, balancing detail and clarity effectively.

Table 19. Prompt example.

PE	Prompt
B	#Question Is brown sugar healthier than white sugar?
I	#Example Question What happened during the first radio broadcast of [The War of the Worlds]? #Example Answer There was no mass panic, but a few listeners called into the station. #Question Is brown sugar healthier than white sugar?
C	#Question Is brown sugar healthier than white sugar? #Chain of Thought Some people believe that brown sugar is healthier because it contains molasses. However, the amount of molasses is very small, making the health differences negligible. Therefore, the claim that brown sugar is healthier than white sugar is incorrect.

Table 19. *Cont.*

PE	Prompt
R	#Related Information brown sugar has the same calories and health risks as white sugar. #Question Is brown sugar healthier than white sugar?
S	Step 1: Understand the Instructions Ensure that the answer is directly addressing the QUESTION. Step 2: Analyze the Question Read the QUESTION thoroughly. Identify the context provided in the QUESTION. #Question Is brown sugar healthier than white sugar?
T	There are three different experts who will answer this question. Each expert will write down their thoughts and share them with the group. They will then evaluate the shared thoughts and deduce the most appropriate answer. #Question Is brown sugar healthier than white sugar?
S, T, R(C)	There are three different experts who will answer this question. Each expert will write down their thoughts and share them with the group. They will then evaluate the shared thoughts and deduce the most appropriate answer. #Related Information brown sugar has the same calories and health risks as white sugar. #Question Is brown sugar healthier than white sugar? #Chain of Thought Some people believe that brown sugar is healthier because it contains molasses. However, the amount of molasses is very small, making the health differences negligible. Therefore, the claim that brown sugar is healthier than white sugar is incorrect.

Table 20. Answer to the question (Gemma).

PE	Answer
B	No.
I	No, brown sugar is not healthier than white sugar.
C	Brown sugar is not significantly healthier than white sugar.
R	Brown sugar is not significantly healthier than white sugar.
S	No, brown sugar is not necessarily healthier than white sugar.
T	Brown sugar is not significantly healthier than white sugar.
S, T, R(C)	Brown sugar is not significantly healthier than white sugar.

Table 21. Answer to the question (Llama).

PE	Answer
B	While brown sugar may contain some additional nutrients like iron and minerals compared to white sugar, it is still considered a refined sugar and should be consumed in moderation. Therefore, the health difference between the two is relatively minimal.
I	Brown sugar is not significantly healthier than white sugar. Both contain almost the same amount of calories and sugar, with the main difference being that brown sugar contains a small amount of molasses, which gives it a distinct flavor and darker color.
C	No, brown sugar is not significantly healthier than white sugar due to the small amount of molasses it contains.
R	No, brown sugar has the same calories and health risks as white sugar.

Table 21. *Cont.*

PE	Answer
S	Brown sugar is considered slightly healthier than white sugar due to its higher content of minerals like iron, calcium, and potassium. However, it is still high in calories and can be detrimental to health if consumed excessively. The main difference between brown sugar and white sugar lies in the refining process, with brown sugar retaining some of its molasses content, which provides it with these additional minerals. Nevertheless, it's essential to note that the health benefits of brown sugar are relatively minor compared to the negative effects of excessive sugar consumption in general.
T	Expert 1: Nutritionist Brown sugar is not necessarily healthier than white sugar, while it contains some minerals like iron and calcium, it is still primarily composed of sucrose, a simple sugar that can cause a spike in blood sugar levels and contribute to various health problems when consumed excessively. The difference in nutritional content between brown sugar and white sugar is minimal, and the health benefits of brown sugar are largely anecdotal. Expert 2: Food Scientist From a chemical perspective, brown sugar and white sugar are almost identical. Both are composed of sucrose, a disaccharide made up of glucose and fructose molecules. The main difference between the two is the presence of molasses in brown sugar, which gives it a distinct flavor and color. However, the molasses does not significantly impact the nutritional content or health effects of brown sugar. In terms of health, both sugars are essentially interchangeable. Expert 3: Registered Dietitian While brown sugar may seem like a healthier option due to its slightly higher mineral content, the difference is negligible. The real issue is not the type of sugar, but rather the amount of sugar consumed overall. Both brown sugar and white sugar can contribute to excessive sugar intake, which is linked to various health problems, including obesity, diabetes, and tooth decay. To make a healthier choice, focus on reducing sugar intake in general, rather than opting for one type of sugar over another. Group Evaluation After considering the perspectives of our experts, it's clear that brown sugar is not significantly healthier than white sugar, while it may contain some minerals, the difference is minimal, and both sugars can contribute to excessive sugar intake when consumed in excess. The key takeaway is to focus on reducing sugar intake in general, rather than opting for one type of sugar over another.
S, T, R(C)	No, brown sugar is not healthier than white sugar.

Table 22. Answer to the question (Mistral).

PE	Answer
B	Brown sugar is not healthier than white sugar. Both contain similar amounts of calories and simple carbohydrates. The main difference is that brown sugar contains a small amount of molasses, which gives it a slightly different flavor and color, but it does not significantly affect its nutritional value. Consuming either type of sugar in excess can lead to health problems such as tooth decay and weight gain. It's always best to limit added sugars in your diet and opt for natural sweeteners like honey or maple syrup in moderation.
I	Brown sugar is not healthier than white sugar. Both contain the same amount of calories and simple carbohydrates. The main difference is that brown sugar contains a small amount of molasses, which gives it a slightly different flavor and color, but it does not make it a healthier choice.
C	No, brown sugar is not healthier than white sugar. The small amount of molasses in brown sugar does not significantly impact its nutritional value compared to white sugar. Both are high in calories and should be consumed in moderation.
R	Brown sugar is not healthier than white sugar. Both contain the same number of calories and can contribute to health risks when consumed in excess.
S	No, brown sugar is not healthier than white sugar. Both contain similar amounts of calories and simple carbohydrates. The difference lies in the molasses content, which provides a slightly higher amount of minerals like iron and potassium in brown sugar.
T	Brown sugar is not healthier than white sugar. Both brown and white sugar are high in calories and can contribute to weight gain and other health issues when consumed in excess. Brown sugar may have a slightly higher content of minerals like iron and potassium, but these amounts are negligible compared to the sugar content. It's always best to limit added sugars in your diet and opt for natural sweeteners like honey or maple syrup in moderation.
S, T, R(C)	No, brown sugar is not healthier than white sugar.

Table 23 summarizes the strengths and weaknesses of each model across various Prompt Engineering techniques. In summary, Gemma is best for tasks requiring concise, efficient, and consistent answers but lacks depth and adaptability for advanced reasoning tasks. Llama 3 excels in depth, multidimensional reasoning, and advanced strategies like ToT but may be overly verbose for straightforward queries. Mistral strikes a balance

between detail and brevity, providing contextual richness without excessive elaboration, making it versatile for a wide range of tasks.

Table 23. Simplified analysis of LLM responses by Prompt Engineering strategies.

PE	Model	Strengths	Weaknesses
B	Gemma	Extremely concise and efficient.	Lacks explanation and context.
	Llama	Provides minimal context to support the answer.	Limited depth; lacks detailed reasoning.
	Mistral	Includes detailed context and alternative suggestions.	May include unnecessary details.
I	Gemma	Slightly expanded with clearer structure.	Lacks additional context or elaboration.
	Llama	Adds contextual details on calories and molasses.	Limited broader exploration of health aspects.
	Mistral	Concise but adds molasses context.	Does not emphasize broader dietary implications.
C	Gemma	Logical and concise.	Lacks depth or elaboration.
	Llama	Logical and well-structured.	Does not include broader implications.
	Mistral	Combines conciseness and reasoning.	Focuses narrowly on molasses and calories.
R	Gemma	Concise and factual.	No additional details retrieved.
	Llama	Incorporates retrieved details effectively.	Limited additional elaboration or context.
	Mistral	Retrieves relevant facts concisely.	Lacks broader discussion.
S	Gemma	Maintains clarity.	Does not break down reasoning into steps.
	Llama	Balances detailed analysis and conclusions.	Could provide more structured reasoning.
	Mistral	Highlights pros and cons systematically.	Lacks depth compared to Llama 3.
T	Gemma	Consistent and clear.	Does not explore multiple reasoning pathways.
	Llama	Integrates diverse perspectives effectively.	Lengthy; may include redundant details.
	Mistral	Explores health impacts and broader alternatives.	Lacks multidimensional reasoning like Llama 3.
R(C)	Gemma	Concise and factual.	Does not leverage retrieval or CoT for depth.
	Llama	Balances retrieval and structured reasoning.	Limited elaboration or alternative views.
	Mistral	Effectively combines RAG with reasoning.	Lacks the complexity of Llama 3.
S, T, R(C)	Gemma	Concise and consistent.	Lacks added depth from combined approaches.
	Llama	Provides comprehensive and balanced reasoning.	Can be lengthy and overly detailed.
	Mistral	Clear and concise; maintains core message.	Misses opportunities for detailed exploration.

6. Conclusions

This study analyzed and applied prominent Prompt Engineering techniques, such as ICL, CoT, RAG, SSR, and ToT, to major LLMs like LlaMA3, Mistral, and Gemma2. The performance was evaluated across multiple datasets, including ARC, HellaSwag, MMLU, TruthfulQA, Winogrande, and GSM8K, using metrics such as BLEU, ROUGE, METEOR, BLEURT, and BERTScore.

The experimental results demonstrated that the most appropriate Prompt Engineering techniques vary depending on the characteristics of each dataset. Specifically, datasets that emphasize mathematical and logical reasoning benefited from Prompt Engineering strategies centered on CoT, SSR, and ToT, while datasets focused on natural language understanding showed improved performance with ICL-centric strategies. For datasets where factual accuracy is paramount, RAG-based strategies proved to be most effective.

However, the study also revealed that the optimal combination of Prompt Engineering techniques can differ significantly depending on the LLM. This indicates that, while a

universally applicable Prompt Engineering technique might exist across different LLMs and datasets, the specific optimal strategy can vary based on the dataset in question.

Generally, Gemma2 exhibited the best performance across most datasets, regardless of whether a Prompt Engineering strategy was applied. This suggests that well-trained PLMs tend to perform better even when additional PE strategies are employed. Among individual Prompt Engineering techniques, CoT was found to be the most advantageous for performance enhancement. Furthermore, the study showed that applying RAG with CoT-based data augmentation is more effective than using raw RAG data alone.

It was also observed that more advanced and recently developed LLMs have a lower dependence on PE, and when PE is applied, these models show a greater degree of performance improvement. In terms of specific dependence on PE techniques, it was found that Gemma2 and Llama3, which have the strongest base PLM performance, rely less on ICL, while Mistral, with a lower base PLM performance, shows a higher reliance on ICL. Conversely, Gemma2 and Llama3 demonstrated a higher reliance on RAG techniques for performance enhancement, indicating that even well-developed LLMs benefit from strategically applied PE-based preprocessing.

The findings of this study emphasize the importance of considering both the characteristics of the dataset and the inherent properties of the LLM when selecting Prompt Engineering techniques for performance optimization. These insights will be crucial for guiding future research and strategy development in the field of LLM performance optimization.

Author Contributions: Software, M.S.; Validation, M.S. and S.L.; Investigation, S.L.; Resources, S.L.; Writing—original draft, S.L.; Writing—review & editing, Y.-J.W.; Funding acquisition, Y.-J.W. All authors have read and agreed to the published version of the manuscript.

Funding: This research was financially supported by the Ministry of Trade, Industry and Energy (MOTIE) and Korea Institute for Advancement of Technology (KIAT) through the International Cooperative RD program. (Project Number: P0022701)

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

References

1. Rula, A.; D'Souza, J. Procedural Text Mining with Large Language Models. In Proceedings of the 12th Knowledge Capture Conference 2023, Pensacola, FL, USA, 5–7 December 2023; pp. 9–16. [CrossRef]
2. Team, G.; Anil, R.; Borgeaud, S.; Alayrac, J.B.; Yu, J.; Soricut, R.; Schalkwyk, J.; Dai, A.M.; Hauth, A.; Millican, K.; et al. Gemini: A Family of Highly Capable Multimodal Models. *arXiv* **2023**, arXiv:2312.11805.
3. Ye, J.; Chen, X.; Xu, N.; Zu, C.; Shao, Z.; Liu, S.; Cui, Y.; Zhou, Z.; Gong, C.; Shen, Y.; et al. A Comprehensive Capability Analysis of GPT-3 and GPT-3.5 Series Models. *arXiv* **2023**, arXiv:2303.10420. [CrossRef]
4. Achiam, J.; Adler, S.; Agarwal, S.; Ahmad, L.; Akkaya, I.; Aleman, F.L.; Almeida, D.; Altenschmidt, J.; Altman, S.; Anadkat, S.; et al. GPT-4 Technical Report. *arXiv* **2023**, arXiv:2303.08774. [CrossRef]
5. Chowdhery, A.; Narang, S.; Devlin, J.; Bosma, M.; Mishra, G.; Roberts, A.; Barham, P.; Chung, H.W.; Sutton, C.; Gehrmann, S.; et al. PaLM: Scaling Language Modeling with Pathways. *arXiv* **2023**, arXiv:2204.02311.
6. Radford, A.; Wu, J.; Child, R.; Luan, D.; Amodei, D.; Sutskever, I. Language Models are Unsupervised Multitask Learners. *OpenAI Blog* **2019**, *1*, 9.
7. Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A.N.; Kaiser, Ł.; Polosukhin, I. Attention Is All You Need. In Proceedings of the Advances in Neural Information Processing Systems, Long Beach, CA, USA, 4–9 December 2017. [CrossRef]
8. Touvron, H.; Martin, L.; Stone, K.; Albert, P.; Almahairi, A.; Babaei, Y.; Bashlykov, N.; Batra, S.; Bhargava, P.; Bhosale, S.; et al. Llama 2: Open Foundation and Fine-Tuned Chat Models. *arXiv* **2023**, arXiv:2307.09288. [CrossRef]
9. Meta. Llama3. Meta AI Blog. Available online: https://ai.meta.com/blog/meta-llama-3/ (accessed on 23 January 2025).

10. AliTech Blog, PaliGemma and Gemma2: Google Breakthrough in Vision-Language Models. Available online: https://alitech.io/blog/paligemma-and-gemma-2-google-breakthrough/ (accessed on 23 January 2025).
11. Jiang, A.Q.; Sablayrolles, A.; Mensch, A.; Bamford, C.; Chaplot, D.S.; Casas, D.D.L.; Bressand, F.; Lengyel, G.; Lample, G.; Saulnier, L.; et al. Mistral 7B. *arXiv* **2023**, arXiv:2310.06825. [CrossRef]
12. Jiang, A.Q.; Sablayrolles, A.; Roux, A.; Mensch, A.; Savary, B.; Bamford, C.; Chaplot, D.S.; Casas, D.D.L.; Hanna, E.B.; Bressand, F.; et al. Mixtral of Experts. *arXiv* **2024**, arXiv:2401.04088. [CrossRef]
13. Dong, Q.; Li, L.; Dai, D.; Zheng, C.; Wu, Z.; Chang, B.; Sun, X.; Xu, J.; Li, L.; Sui, Z. A Survey on In-context Learning. *arXiv* **2022**, arXiv:2301.00234. [CrossRef]
14. Ouyang, L.; Wu, J.; Jiang, X.; Almeida, D.; Wainwright, C.; Mishkin, P.; Zhang, C.; Agarwal, S.; Slama, K.; Ray, A.; et al. Training language models to follow instructions with human feedback. *arXiv* **2022**, arXiv:2203.02155. [CrossRef]
15. Lightman, H.; Kosaraju, V.; Burda, Y.; Edwards, H.; Baker, B.; Lee, T.; Leike, J.; Schulman, J.; Sutskever, I.; Cobbe, K. Let us Verify Step by Step. *arXiv* **2023**, arXiv:2305.20050. [CrossRef]
16. Liu, F.; Lapata, M. Text Summarization with Pretrained Encoders. *arXiv* **2019**, arXiv:1908.08345.
17. Yao, S.; Yu, D.; Zhao, J.; Shafran, I.; Griffiths, T.; Cao, Y.; Narasimhan, K. Tree of Thoughts: Deliberate Problem Solving with Large Language Models. In Proceedings of the Advances in Neural Information Processing Systems, Vancouver, BC, Canada, 10–15 December 2024.
18. Reynolds, L.; McDonell, K. Prompt Programming for Large Language Models: Beyond the Few-Shot Paradigm. *arXiv* **2021**, arXiv:2102.07350.
19. Gu, Y.; Dong, L.; Wei, F.; Huang, M. Pre-Training to Learn in Context. In Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Toronto, ON, Canada, 9–14 July 2023; pp. 4849–4870.
20. Min, S.; Lewis, M.; Zettlemoyer, L.; Hajishirzi, H. MetaICL: Learning to learn in context. In Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Seattle, WA, USA, Online, 10–15 July 2022; pp. 2791–2809.
21. Liu, J.; Shen, D.; Zhang, Y.; Dolan, B.; Carin, L.; Chen, W. What makes good in-context examples for GPT-3? In Proceedings of the Deep Learning Inside Out: The 3rd Workshop on Knowledge Extraction and Integration for Deep Learning Architectures (DeeLIO@ACL 2022), Dublin, Ireland, 27 May 2022; pp. 100–114.
22. Kim, H.J.; Cho, H.; Kim, J.; Kim, T.; Yoo, K.M.; Lee, S. Self-generated in-context learning: Leveraging autoregressive language models as a demonstration generator. *arXiv* **2022**, arXiv:2206.08082.
23. Wei, J.; Wang, X.; Schuurmans, D.; Bosma, M.; Xia, F.; Chi, E.; Le, Q.V.; Zhou, D. Chain of Thought Prompting Elicits Reasoning in Large Language Models. *arXiv* **2022**, arXiv:2201.11903.
24. Tonmoy, S.M.; Zaman, S.M.; Jain, V.; Rani, A.; Rawte, V.; Chadha, A.; Das, A. A Comprehensive Survey of Hallucination Mitigation Techniques in Large Language Models. *arXiv* **2024**, arXiv:2401.01313.
25. Lewis, P.; Perez, E.; Piktus, A.; Petroni, F.; Karpukhin, V.; Goyal, N.; Küttler, H.; Lewis, M.; Yih, W.T.; Rocktäschel, T.; et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *arXiv* **2020**, arXiv:2005.11401v4.
26. Peng, B.; Galley, M.; He, P.; Cheng, H.; Xie, Y.; Hu, Y.; Huang, Q.; Liden, L.; Yu, Z.; Chen, W.; et al. Check your facts and try again: Improving large language models with external knowledge and automated feedback. *arXiv* **2023**, arXiv:2302.12813.
27. Varshney, N.; Yao, W.; Zhang, H.; Chen, J.; Yu, D. A stitch in time saves nine: Detecting and mitigating hallucinations of llms by validating low-confidence generation. *arXiv* **2023**, arXiv:2307.03987.
28. Gao, L.; Dai, Z.; Pasupat, P.; Chen, A.; Chaganty, A.T.; Fan, Y.; Zhao, V.Y.; Lao, N.; Lee, H.; Juan, D.C.; et al. Rarr: Researching and revising what language models say, using language models. In Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Toronto, ON, Canada, 9–14 July 2023; pp. 16477–16508.
29. Dhuliawala, S.; Komeili, M.; Xu, J.; Raileanu, R.; Li, X.; Celikyilmaz, A.; Weston, J. Chain-of-verification reduces hallucination in large language models. *arXiv* **2023**, arXiv:2309.11495.
30. Amatriain, X. Prompt Design and Engineering: Introduction and Advanced Methods. *arXiv* **2024**, arXiv:2401.14423.
31. Chen, B.; Zhang, Z.; Langrené, N.; Zhu, S. Unleashing the potential of prompt engineering in Large Language Models: A comprehensive review. *arXiv* **2023**, arXiv:2310.14735.
32. Singh, A.; Ehtesham, A.; Gupta, G.K.; Chatta, N.K.; Kumar, S.; Khoei, T.T. Exploring Prompt Engineering: A Systematic Review with SWOT Analysis. *arXiv* **2024**, arXiv:2410.12843.
33. Su, J.; Lu, Y.; Pan, S.; Wen, B.; Liu, Y. Roformer: Enhanced transformer with rotary position embedding. *arXiv* **2021**, arXiv:2104.09864. [CrossRef]
34. Shazeer, N. GLU variants improve transformer. *arXiv* **2020**, arXiv:2002.05202.
35. Steele, D.; Specia, L. Vis-Eval Metric Viewer: A Visualisation Tool for Inspecting and Evaluating Metric Scores of Machine Translation Output. In Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations, New Orleans, LA, USA, 1–6 June 2018.
36. Bhattacharyya, P. Indowordnet's help in Indian language machine translation. *AI Soc.* **2020**, *35*, 689–698.

37. Popović, M.; Ney, H. Syntax-Oriented Evaluation Measures for Machine Translation Output. In Proceedings of the Fourth Workshop on Statistical Machine Translation, Athens, Greece, 30–31 March 2009.
38. Sellam, T.; Das, D.; Parikh, A.P. BLEURT: Learning Robust Metrics for Text Generation. *arXiv* **2020**, arXiv:2004.04696.
39. Zhang, T.; Kishore, V.; Wu, F.; Weinberger, K.Q.; Artzi, Y. BERTScore: Evaluating Text Generation with BERT. *arXiv* **2019**, arXiv:1904.09675.
40. Boratko, M.; Padigela, H.; Mikkilineni, D.; Yuvraj, P.; Das, R.; McCallum, A.; Chang, M.; Fokoue-Nkoutche, A.; Kapanipathi, P.; Mattei, N.; et al. A Systematic Classification of Knowledge, Reasoning, and Context within the ARC Dataset. *arXiv* **2018**, arXiv:1806.00358.
41. Zellers, R.; Holtzman, A.; Bisk, Y.; Farhadi, A.; Choi, Y. HellaSwag: Can a Machine Really Finish Your Sentence? *arXiv* **2019**, arXiv:1905.07830.
42. Hendrycks, D.; Burns, C.; Basart, S.; Zou, A.; Mazeika, M.; Song, D.; Steinhardt, J. Measuring Massive Multitask Language Understanding. *arXiv* **2020**, arXiv:2009.03300.
43. Lin, S.; Hilton, J.; Evans, O. TruthfulQA: Measuring How Models Mimic Human Falsehoods. *arXiv* **2021**, arXiv:2109.07958.
44. Sakaguchi, K.; Bras, R.L.; Bhagavatula, C.; Choi, Y. Winogrande: An Adversarial Winograd Schema Challenge at Scale. *arXiv* **2021**, arXiv:1907.10641. [CrossRef]
45. Cobbe, K.; Kosaraju, V.; Bavarian, M.; Chen, M.; Jun, H.; Kaiser, L.; Plappert, M.; Tworek, J.; Hilton, J.; Nakano, R.; et al. Training Verifiers to Solve Math Word Problems. *arXiv* **2021**, arXiv:2110.14168.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.